



UNIVERSITÀ
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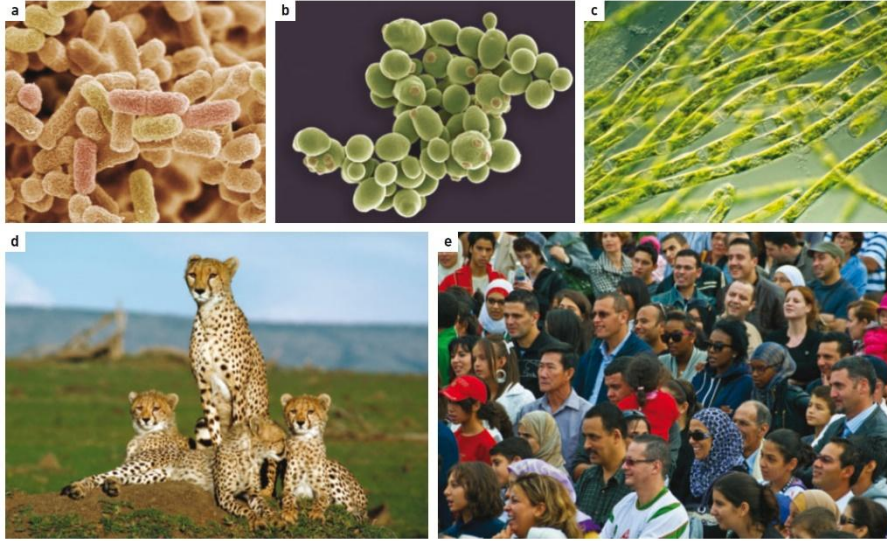
The Cell

MATTEO DIANO, PH.D.

COMPUTATIONAL APPLICATION TO COGNITIVE NEUROSCIENCE

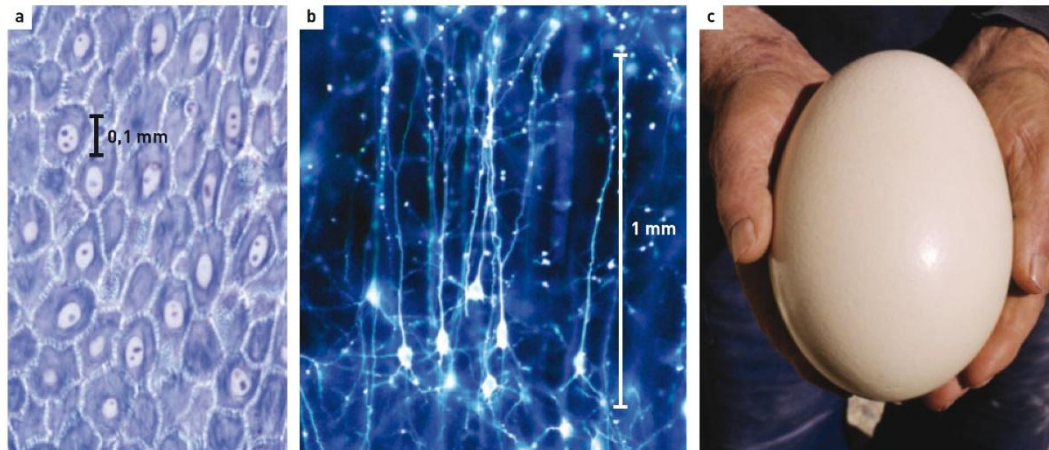
The Cell

FIGURA 1.10 Organismi unicellulari e pluricellulari. Tutti gli organismi sono costituiti da cellule: (a) batteri; (b) lievito di birra; (c) alghe; (d) ghepardi; (e) esseri umani.



Every known organism is made up of either a single cell or a collection of a few or many cells.

FIGURA 1.11 La varietà delle cellule. Le cellule variano notevolmente per dimensioni e forma: (a) cellule della pelle; (b) cellule nervose; (c) uovo di struzzo.



Cells can become specialized and, in some cases, very large.

ESSENTIAL CHARACTERISTICS OF A CELL:

- **Nucleic acids (DNA–RNA)** store and transmit the information needed for a cell's growth, functioning, and reproduction.
- **Membranes** enclose cells and their internal environment.
- **Metabolism** converts energy from the environment into a form that cells can use.

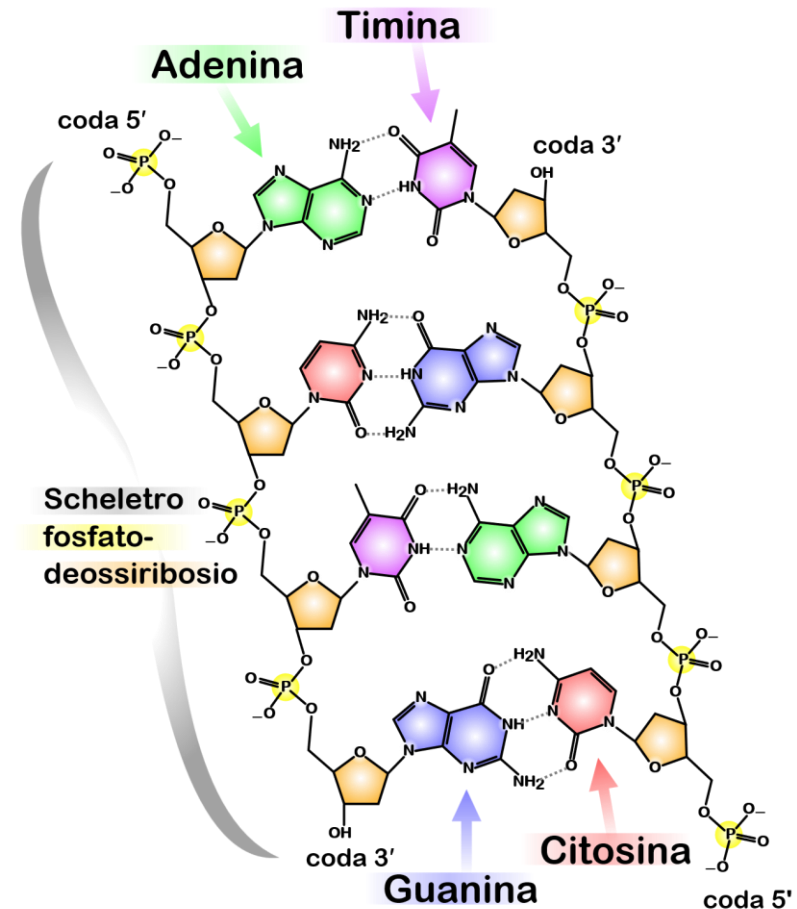


ESSENTIAL CHARACTERISTICS OF A CELL:

1) Nucleic acids (DNA–RNA) store and transmit the information necessary for the cell's growth, functioning, and reproduction.

- **Ability to store and transmit information** (the presence of a stable information archive that encodes their physical characteristics)
- **Ability to reproduce**
- **The subunits represent a four-letter alphabet** that guides the production of proteins

Information archive Deoxyribonucleic Acid—DNA)



ESSENTIAL CHARACTERISTICS OF A CELL:

How do the instructions in DNA direct protein synthesis?

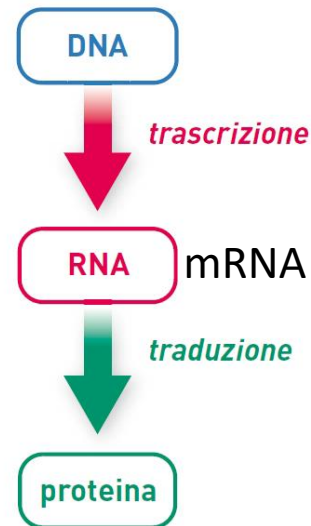


FIGURA 1.13
Il dogma centrale della biologia molecolare. In tutti gli esseri viventi, il dogma centrale indica la direzione del flusso di informazioni dal DNA all'RNA alle proteine.

TRANSCRIPTION: the process in which information is copied from one form to another.

TRANSLATION: the process of converting information stored in the language of nucleic acids (mRNA) into the language of proteins.

GENES: specific DNA sequences that correspond to and contribute to encoding a specific protein product.

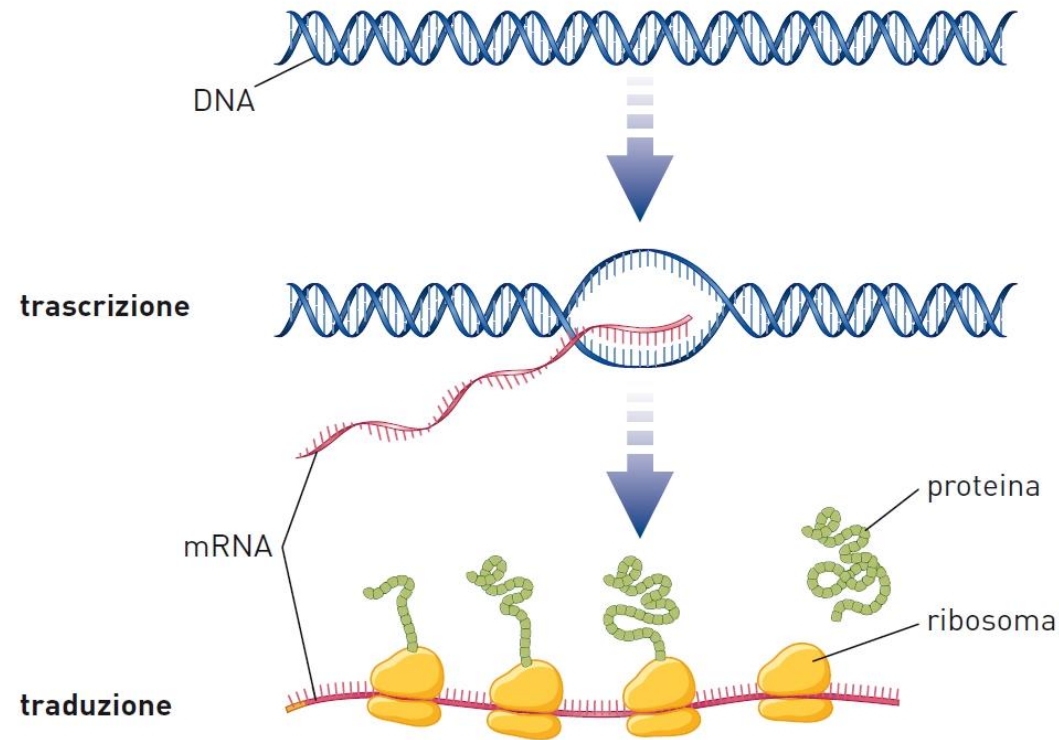
REPLICATION: beyond being preserved, information can also be replicated, copied, allowing genetic information to pass from one cell to another.

MUTATIONS: changes in the DNA sequence.

TRANSCRIPTION: the process in which the genetic information contained in DNA is used as a template to synthesize a messenger RNA (mRNA) molecule.

FIGURA 3.11 Il dogma centrale della biologia molecolare.

Il normale flusso di informazioni in una cellula va dal DNA all'RNA alle proteine.



Oltre a svolgere diverse funzioni essenziali nella cellula, le proteine hanno anche un ruolo strutturale.

The second step in the processing of genetic information is **TRANSLATION**.

Transcription (DNA → primary RNA): RNA polymerase binds a promoter, reads the DNA template, and makes a complementary RNA transcript until a termination sequence.

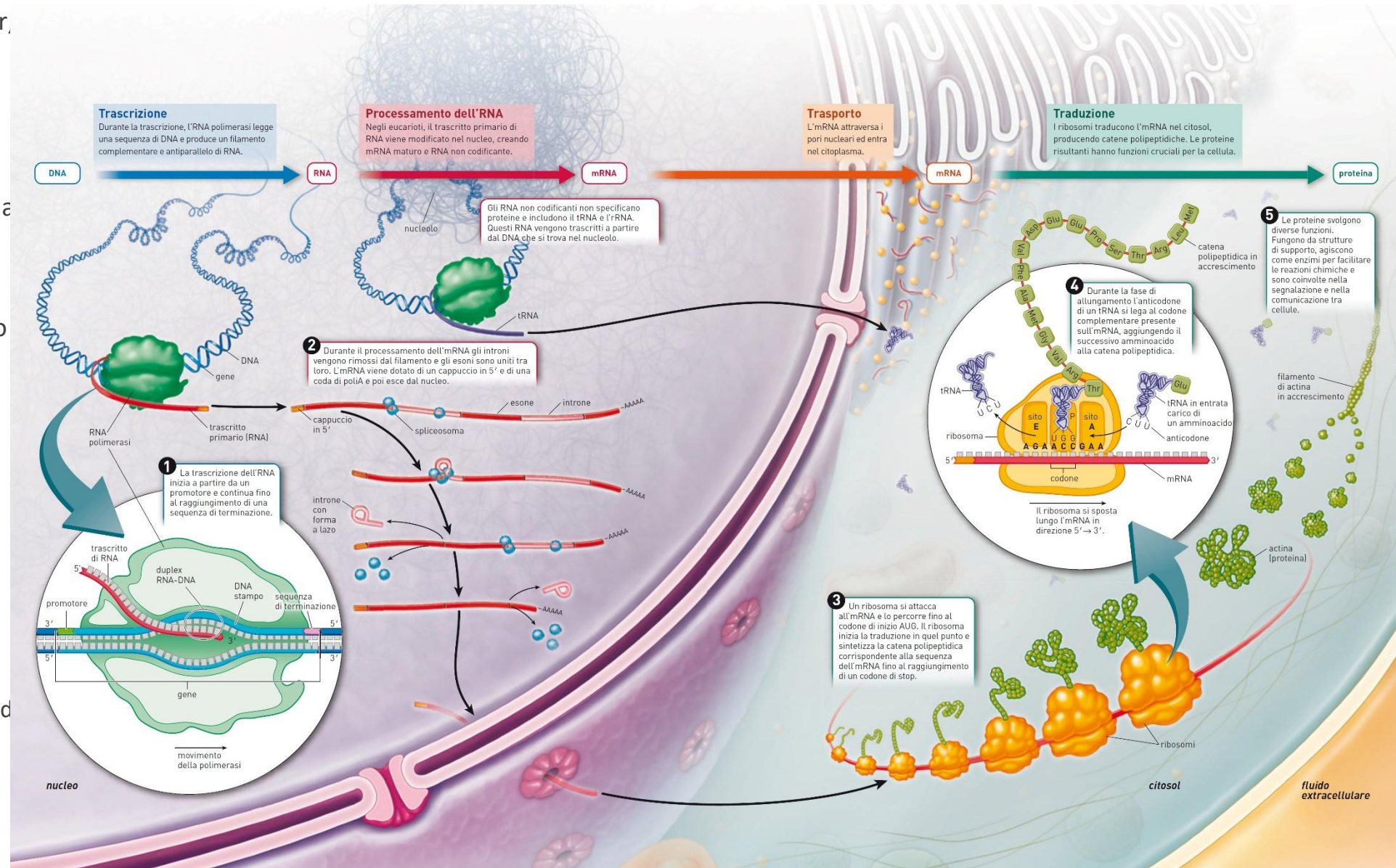
RNA processing (eukaryotes): The pre-mRNA gets a 5' cap, 3' poly-A tail, and introns are spliced out by the spliceosome to join exons → mature mRNA. (Other non-coding RNAs like rRNA and tRNA are also transcribed.)

Nuclear export: Mature mRNA passes through a nuclear pore into the cytosol.

Translation initiation: A ribosome binds the mRNA (near the 5' end), finds the start codon (AUG), and begins protein synthesis.

Elongation & termination: tRNAs match codons with anticodons to add amino acids; the ribosome moves 5'→3' along the mRNA (A/P/E sites). At a stop codon, the polypeptide is released.

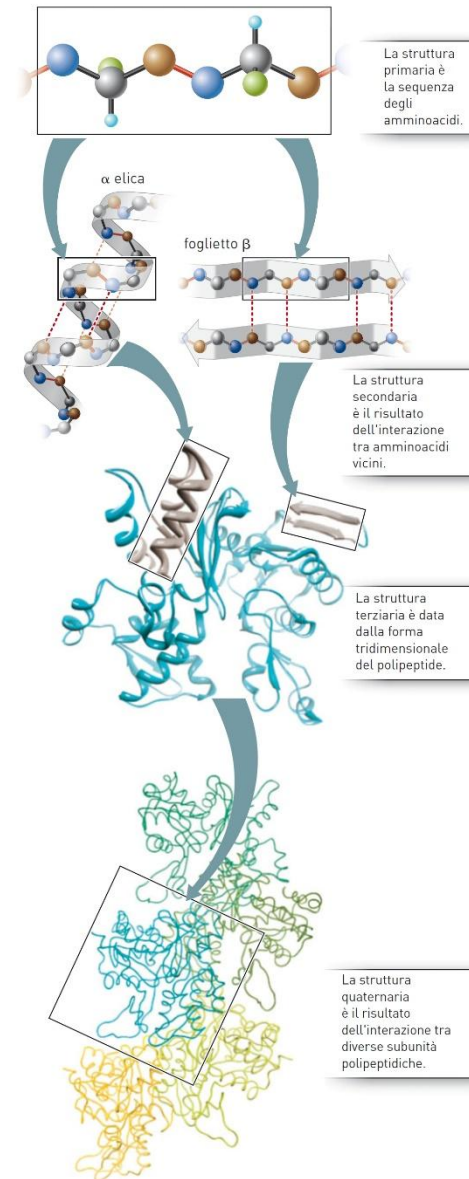
Protein folding & function: The polypeptide folds (and may be modified), forming functional proteins that act in structure, catalysis, signaling, etc



STRUCTURE OF THE PROTEINS

THE AMINO ACID SEQUENCE DETERMINES A PROTEIN'S FOLDING, AND ITS FUNCTION DEPENDS ON THAT FOLD.

- **Primary:** linear amino-acid sequence (N→C); peptide bonds; sets all higher structure.
- **Secondary:** local motifs— α -helices, β -sheets, turns; stabilized by backbone H-bonds.
- **Tertiary:** 3D fold of one chain; hydrophobic core + side-chain H-bonds/ionic/disulfide; forms active sites.
- **Quaternary:** assembly of multiple subunits; homo/hetero; enables cooperativity (e.g., hemoglobin).

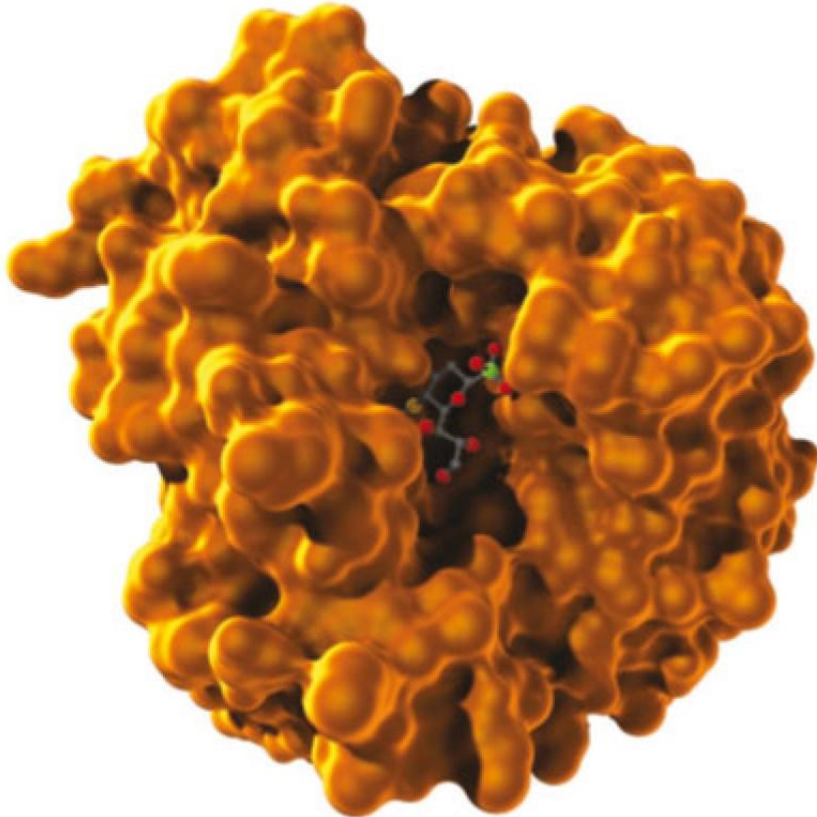


STRUCTURE OF THE PROTEINS

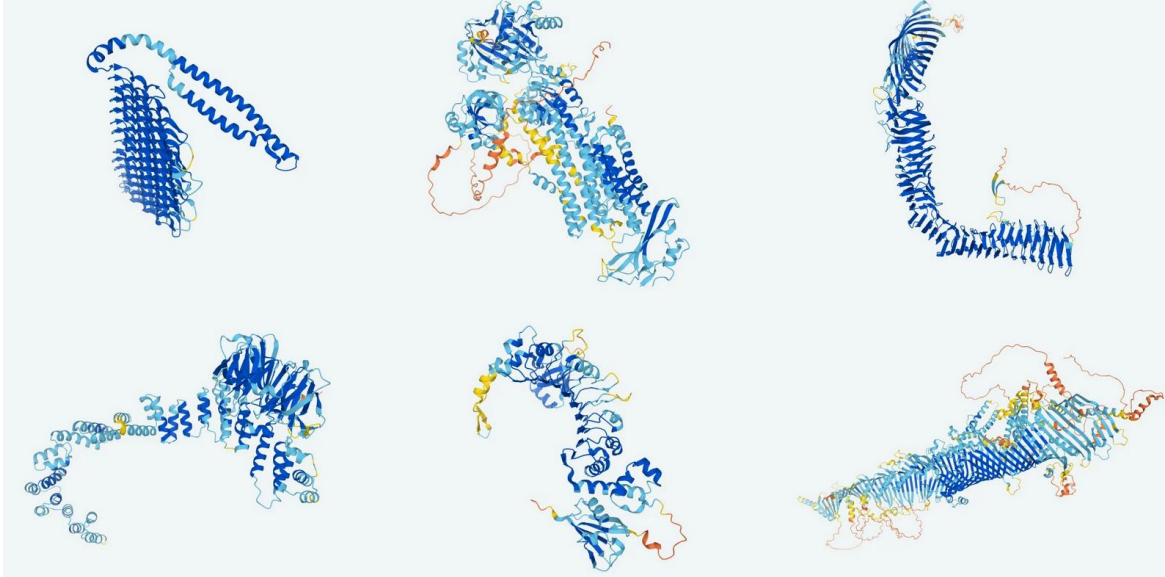
THE AMINO ACID SEQUENCE DETERMINES A PROTEIN'S FOLDING, AND ITS FUNCTION DEPENDS ON THAT FOLD.

FIGURA 4.8 La struttura terziaria determina la funzione.

Questa proteina batterica ha una cavità che può legare una piccola molecola (mostrata al centro con un modello a sfere e bastoncini).



The folding, and therefore the functionality, of a protein can be compromised by chemicals or high temperatures, a process known as **denaturation**.



CRISPR system Cascade subunit CasA, Sequence length 502

[Copy sequence](#)

1

MNLLIDNWIPVRPRNGGKVQIIINLQSLYCSRDQWRLSLPRDDMELAAALALLVCIGI IAPAKDDVEFHRIMNPLTEDEFOQLIAPWIDMFYLNHAEHPFMQTKGVKANDVTMEKLLAGVSGATNCAFVNQPGQGEALCGGCTAIALFNOANQA

161

171

PGFGGGFKSGLRGSTPVTTFVRGIDRSTVLNLVLTPLRQKQFPNESHTENQPTWIKPKISNESIPASSIGFVRGLFWQPAHIELCDPIGIGKSCCGQESNLRYTGFLKEKFTFTVNGLWPHPHSPCLVTVKGEVEEKFLAFTTSAPSWTQI

321

331

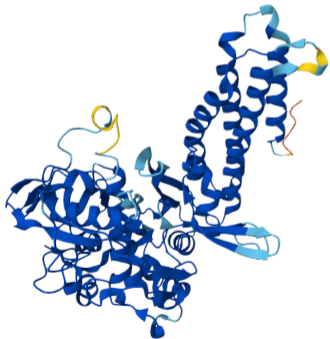
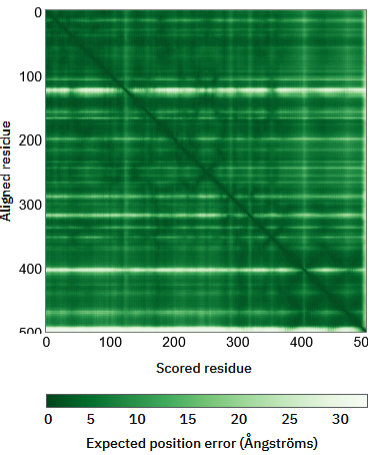
SRVVVDKI IQNENGNNRVAAVNQFRNIAPQSPLELIMGGYRNQASILERRHDVLMFNQGWQQYGNVINEIVTVGLGYKTALRKALYTFAGGFKNKDFRGAGVSVHETAERHFYRQSELLIPDVLANVNFSDADEVIALDLRLHQLCEMLFNQS

481

491

VAPYAHHPKLISLTALARATLYKHLRELKPQGGPSNG

501



Model Confidence

Very high (pLDDT > 90)

High (90 > pLDDT > 70)

Low (70 > pLDDT > 50)

Very low (pLDDT < 50)

pLDDT is a per-residue measure of local confidence. [Learn more...](#)

Domains (

Annotation

D(4) dopamine receptor, Sequence length 419

[Copy sequence](#)

1

MGNRSTADADGLLACRGPAAGASAGASAGLAGOGAAALVGGVLLIGAVLAGNSLVCSVATERALQTPTNSPFVSLAAADLLALLVLPLFVYSEVQGGANLLSPRLCDALMAMDVMLCTASIFNLCAISVDRFVAVAVPLRYNRQGGSRRLQLL

161

171

IGATWLLSAAVAAFVLCGLNDVVRGRDPFAVCRLEDRDYVYVSVCSFFLPCPLMLLLYWATFRGLQRWEVARRAKLHGRAPRRPSGPGPPSPPTPPAPRLPQDPCGPDCAFPAPGLPRGPGCDCAFAAFLPQDPCGPDCAFPAPGLPFPDPCGSNC

321

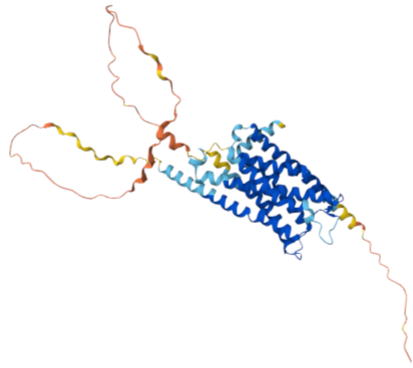
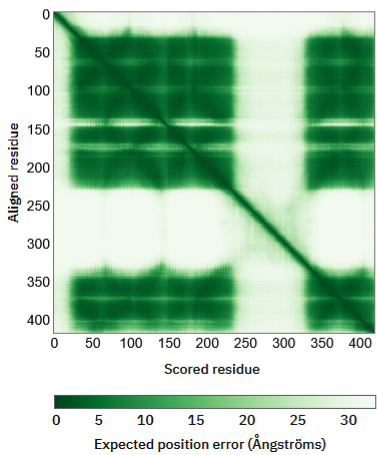
331

APPDAVRAAALPPQTPQTRRRRRRAKITGREKAMRVLPVVVGAFLLCWTFFVWHITQALCPACSVFPRLVSAVTWLGYNLSALNFVIYTVFNAEFRNVFKALRACC

481

491

501



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Domains (1)

Annotations

STRUCTURE OF THE PROTEINS

THE AMINO ACID SEQUENCE DETERMINES A PROTEIN'S FOLDING, AND ITS FUNCTION DEPENDS ON THAT FOLD.

QUATERNARY STRUCTURE

Polypeptide subunits can join to form quaternary structures.

The polypeptide subunits can be identical or different.

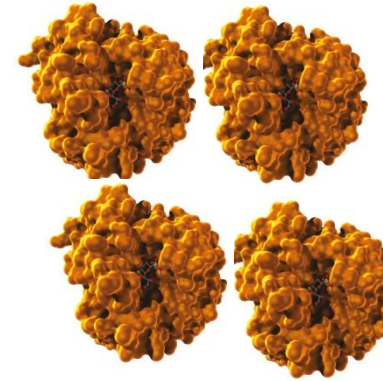
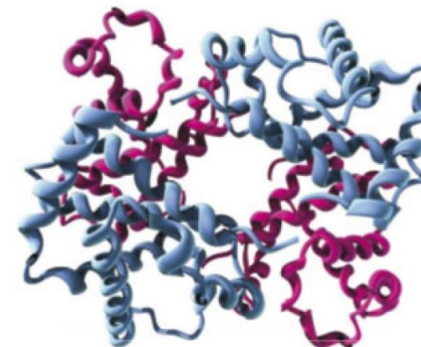


FIGURA 4.10 La struttura quaternaria. Le subunità polipeptidiche delle proteine possono essere identiche, come (a) in un enzima dell'HIV, o diverse, come (b) nell'emoglobina.



Questo enzima è costituito da due subunità polipeptidiche identiche, rappresentate in verde chiaro e verde scuro.



L'emoglobina è composta da quattro subunità polipeptidiche uguali a due a due, rappresentate in viola e blu.

FUNCTIONS OF THE PROTEINS

Proteins carry out diverse cellular functions that depend on their 3D structure:

- Some act as structural elements (e.g., microtubules/filaments).
- Some mediate communication across compartments or membranes (e.g., channels).
- Some speed up chemical reactions (e.g., enzymes).

FIGURA 10.17 Il collagene. Le molecole di collagene di tipo I sono organizzate in una tripla elica e raggruppate in fasci, chiamati fibrille, che a loro volta sono riuniti in fibre. Questo tipo di disposizione, che si osserva nelle fibre tessili e nei cavi d'acciaio, conferisce una forza eccezionale.

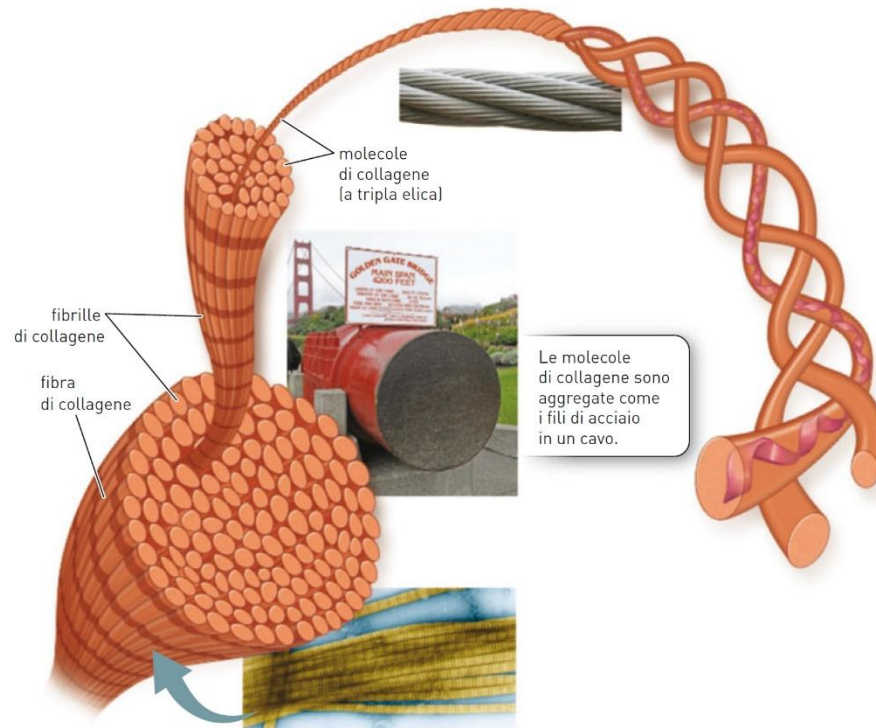
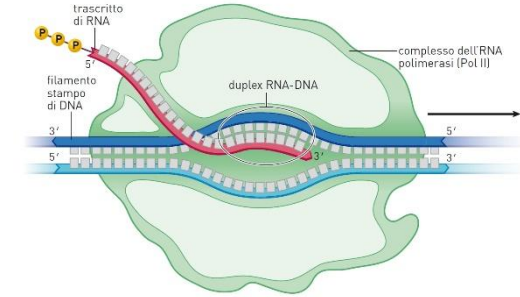
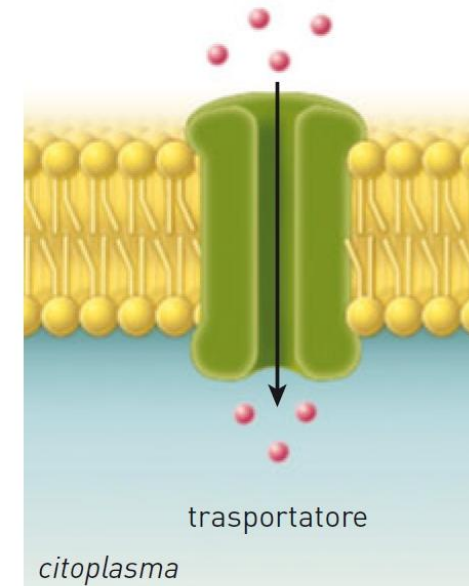


FIGURA 3.16 La bolla di trascrizione. All'interno dell'RNA polimerasi, i due filamenti di DNA si separano e il filamento di RNA in allungamento forma un duplex [doppio filamento] con lo stampo di DNA.



*fluido
extracellulare*



ESSENTIAL CHARACTERISTICS OF A CELL:

2) Membranes enclose cells and their internal environment.

- **CELLULAR MEMBRANE:** separates the living material inside the cell from the surrounding nonliving environment

FIGURA 5.16 Cellule procariote ed eucariote. Le cellule procariote sono prive di un nucleo e di un'estesa compartimentazione interna, caratteristiche che sono invece presenti nelle cellule eucariote.

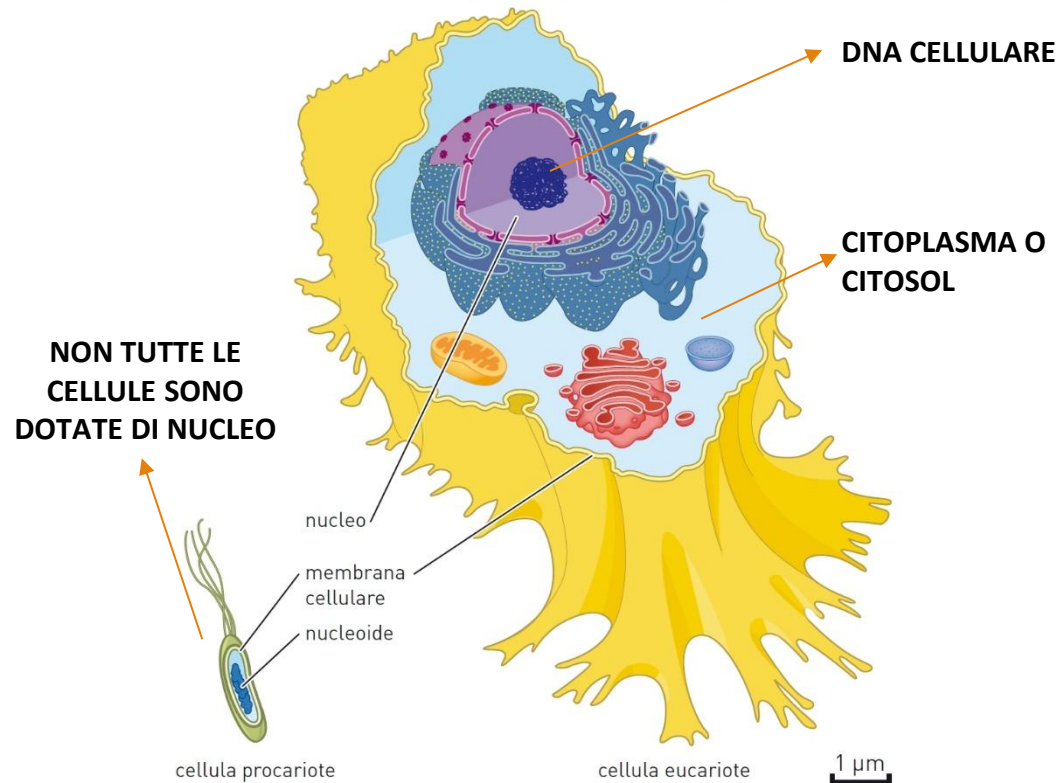
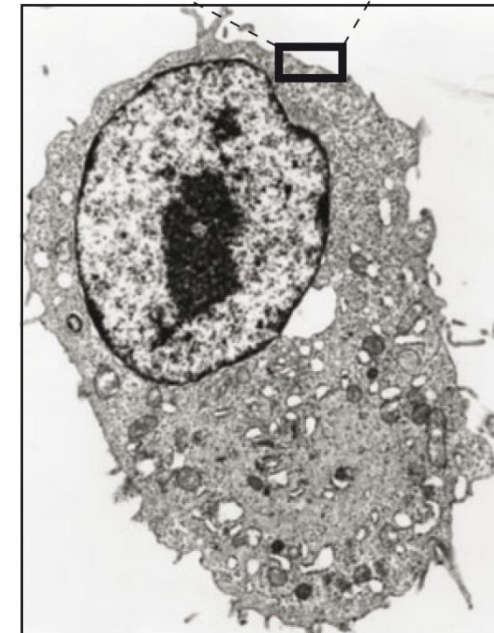
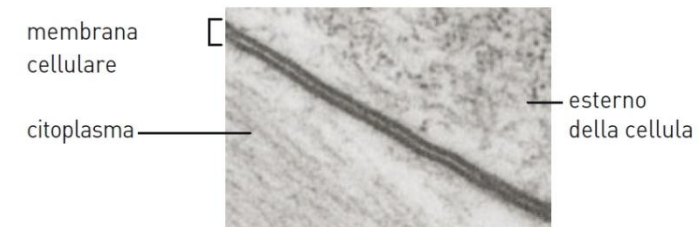


FIGURA 1.14 La membrana cellulare. La membrana cellulare delimita la cellula e controlla gli scambi di sostanze con l'ambiente.



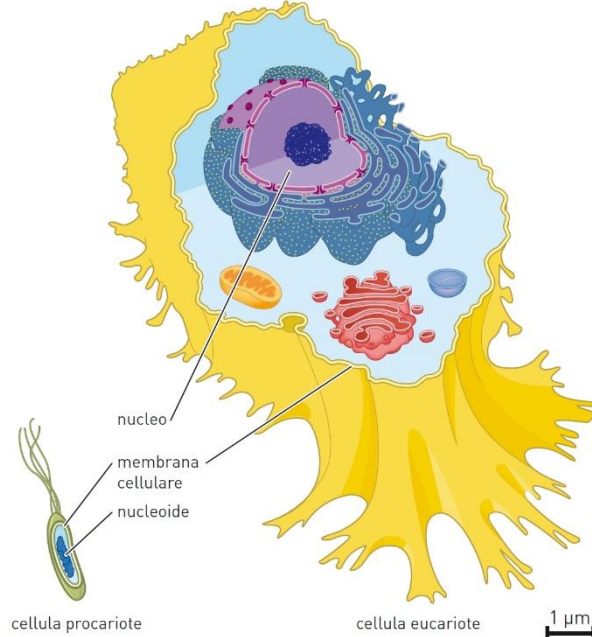
micrografia elettronica a trasmissione di una cellula

ESSENTIAL CHARACTERISTICS OF A CELL:

The presence or absence of a nucleus defines a cell as prokaryotic or eukaryotic.

- Prokaryotic: no nucleus; DNA in a nucleoid.
- Eukaryotic: true nucleus; membrane-bound organelles.

FIGURA 5.16 Cellule procariote ed eucariote. Le cellule procariote sono prive di un nucleo e di un'estesa compartimentazione interna, caratteristiche che sono invece presenti nelle cellule eucariote.



Prokaryotic cell (bacteria).

Eukaryotic cell includes:

animals,
plants,
fungi,
unicellular microorganisms—protists (e.g.,
yeasts).

THE STRUCTURE OF CELL MEMBRANES

Cell membranes are made of a **phospholipid bilayer**.

Lipids with a bulky head and one fatty-acid tail aggregate into a spherical **micelle**.

Lipids with a smaller head and two fatty-acid tails aggregate to form a bilayer.

Liposomes: vesicles composed of phospholipids.

FIGURA 5.2 La struttura di un fosfolipide. I fosfolipidi, i componenti principali delle membrane cellulari, sono costituiti da glicerolo legato a una testa contenente fosfato e a due code di acidi grassi. I fosfolipidi sono anfipatici perché possiedono domini sia idrofili sia idrofobi.

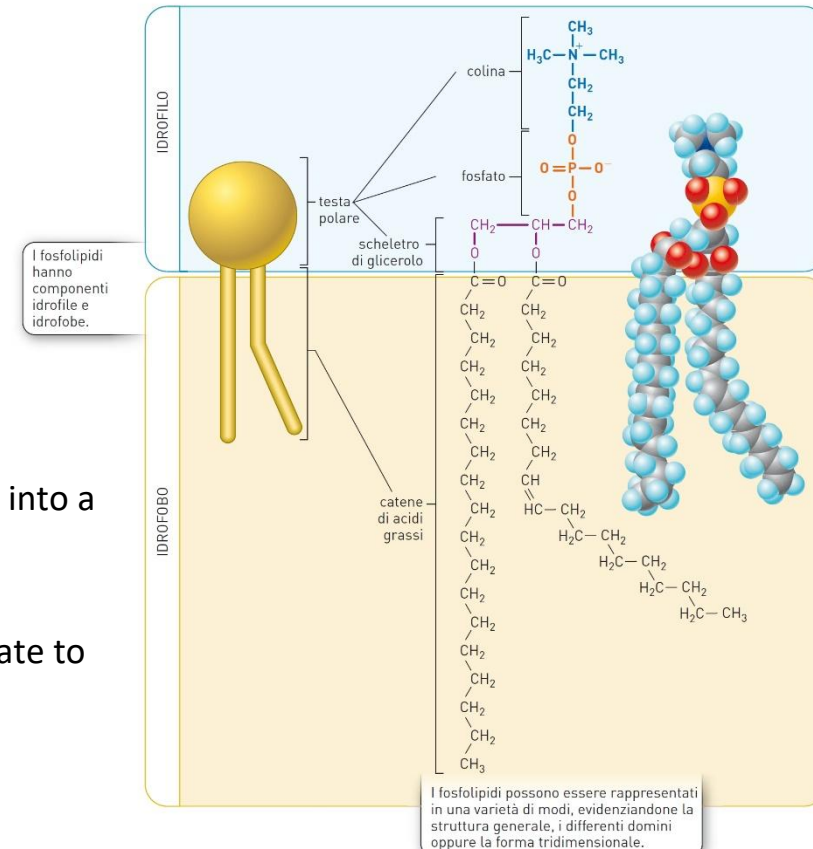
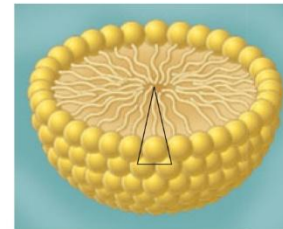
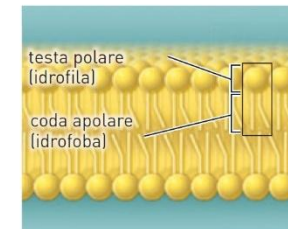


FIGURA 5.3 Diverse strutture create dai fosfolipidi. In ambiente acquoso i fosfolipidi possono formare (a) micelle, (b) doppi strati o (c) liposomi.

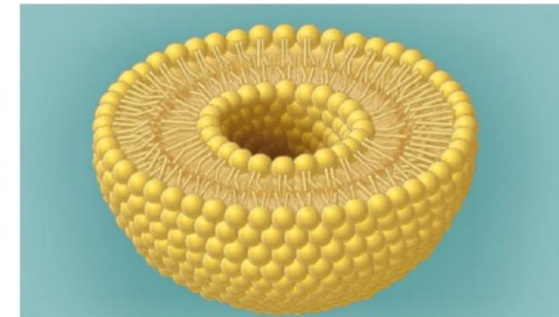
a. micella



b. doppio strato



c. liposoma



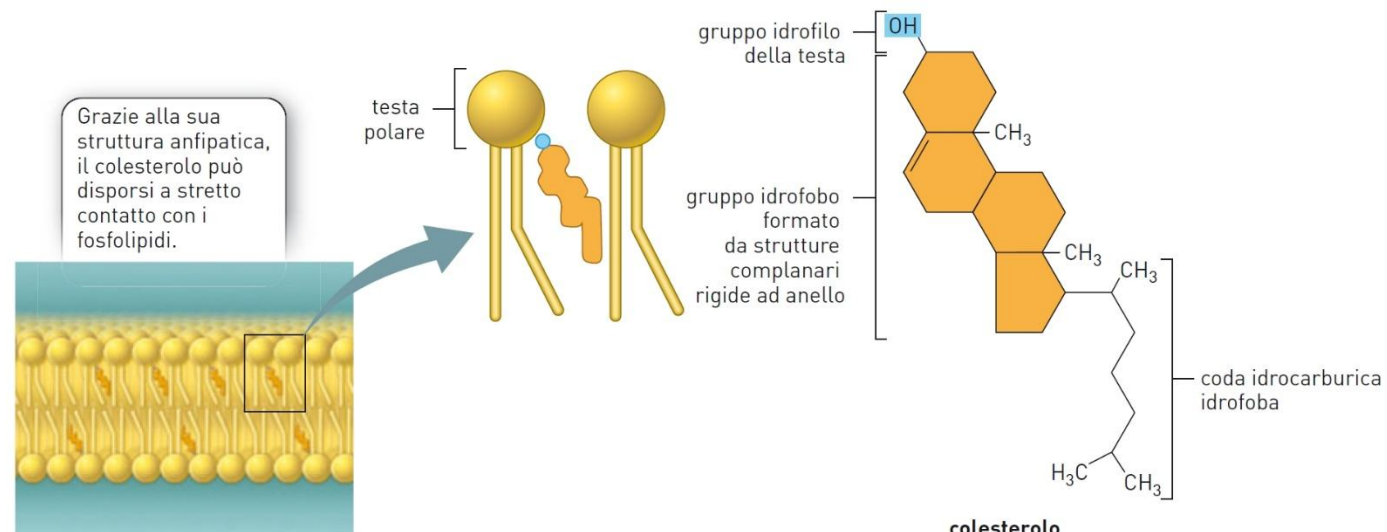
THE STRUCTURE OF CELL MEMBRANES

Cell membranes are dynamic: because lipids can move within the plane of the membrane, **the membrane is said to be fluid.**

Membranes also contain:

- Other lipids such as cholesterol (~30%).
- Proteins (up to ~50% of the mass of a red blood cell membrane).

FIGURA 5.5 Il colesterolo nel doppio strato lipidico. Le molecole di colesterolo incorporate nel doppio strato lipidico influenzano la fluidità della membrana.

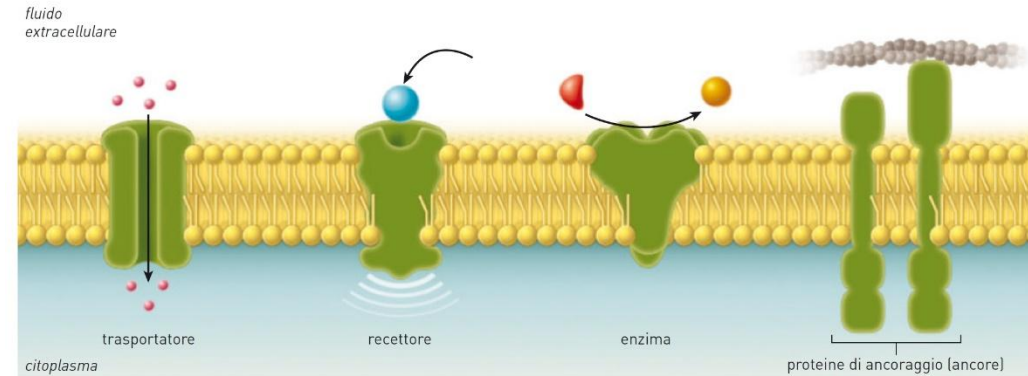


THE STRUCTURE OF CELL MEMBRANES

Proteins associate with cell membranes in different ways and act as:

- **TRANSPORTERS** (MOVE IONS OR OTHER MOLECULES ACROSS THE MEMBRANE)
- **MEMBRANE RECEPTORS** (LET THE CELL RECEIVE SIGNALS FROM THE EXTERNAL ENVIRONMENT)
- **ENZYMES** (CATALYZE CHEMICAL REACTIONS)
- **ANCHORS** (ATTACH TO OTHER PROTEINS AND HELP MAINTAIN THE CELL'S STRUCTURE AND SHAPE)

FIGURA 5.6 Le funzioni delle proteine di membrana.

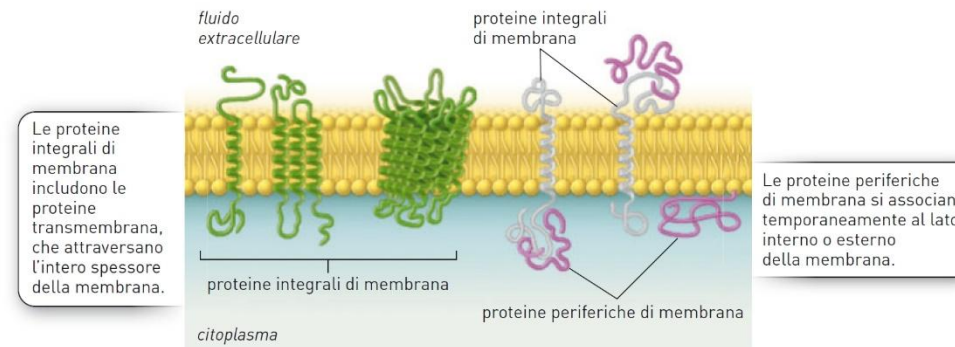


Membrane proteins are also divided into:

Integral (transmembrane) proteins: span the lipid bilayer; tightly embedded (extracted only with detergents). Include channels, transporters, receptors; often α -helices or β -barrels.

Peripheral membrane proteins: attach to the membrane surface via ionic/H-bond interactions or lipid anchors; Often enzymes, adapters, or cytoskeletal anchors.

FIGURA 5.7 Proteine di membrana integrali e periferiche. Le proteine integrali sono associate permanentemente alla membrana. Le proteine periferiche, invece, sono temporaneamente associate a uno dei due strati lipidici o a una proteina integrale.

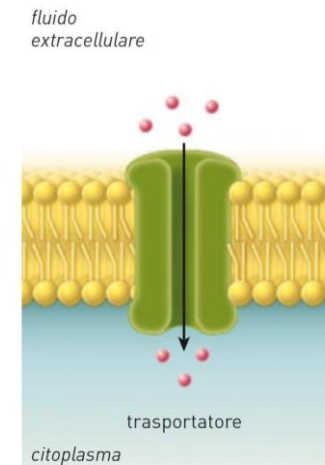
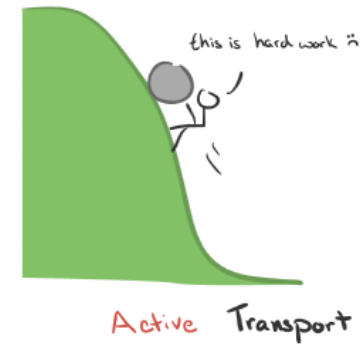
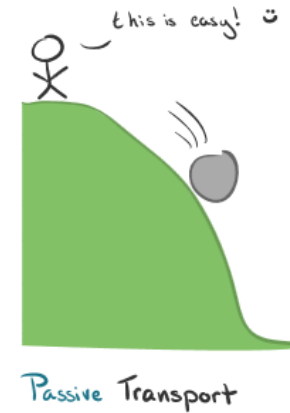


TRANSPORT ACROSS THE CELL MEMBRANE.

- IT CAN BE PASSIVE (NO ENERGY USE) OR ACTIVE (REQUIRES ENERGY).

THE CELL MEMBRANE IS SELECTIVELY PERMEABLE (SEMIPERMEABLE):

- SOME MACROMOLECULES LIKE PROTEINS AND POLYSACCHARIDES ARE TOO LARGE TO CROSS THE MEMBRANE ON THEIR OWN.
- GASES SUCH AS OXYGEN AND CARBON DIOXIDE, AND NONPOLAR MOLECULES LIKE LIPIDS, CAN CROSS THE MEMBRANE.
- SMALL POLAR MOLECULES: WATER CROSSES (ESPECIALLY VIA AQUAPORINS); IONS REQUIRE CHANNELS/TRANSPORTERS AND DON'T CROSS FREELY.

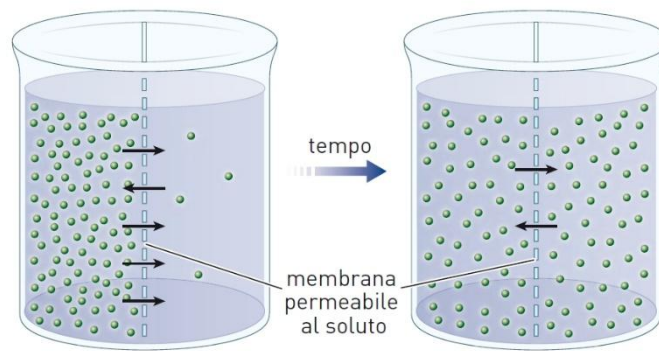


PASSIVE TRANSPORT

The simplest movement between the inside and outside of the cell is passive transport (no energy use).

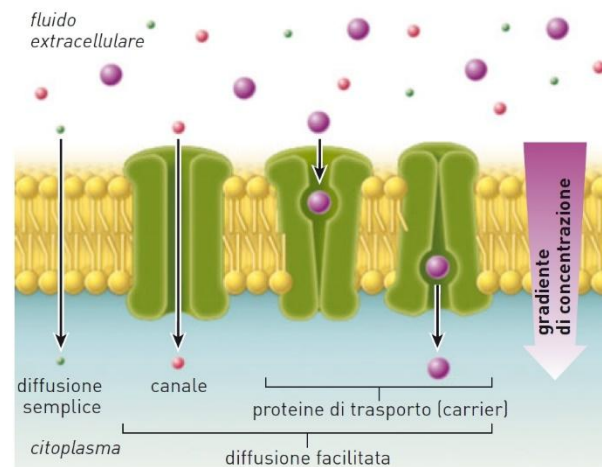
- It occurs by **diffusion**, which relies on the **random motion of molecules** and proceeds **down a concentration gradient**.

FIGURA 5.9 La diffusione è lo spostamento delle molecole dovuto al loro moto casuale. Quando esistono differenze di concentrazione, si verifica un movimento netto delle molecole.



maggiore concentrazione di soluto	minore concentrazione di soluto
Si verifica un movimento netto di soluto dall'area in cui è più concentrato a quella in cui è meno concentrato.	Non c'è nessun movimento netto, ma le molecole di soluto continuano a spostarsi in entrambe le direzioni.

FIGURA 5.10 Diffusione semplice e diffusione facilitata attraverso la membrana cellulare.



DIFFUSIONE SEMPLICE (O₂)

DIFFUSIONE FACILITATA
(HA BISOGNO DI UN TRASPORTATORE)

ACQUAPORINE: TRASPORTATORI DI ACQUA

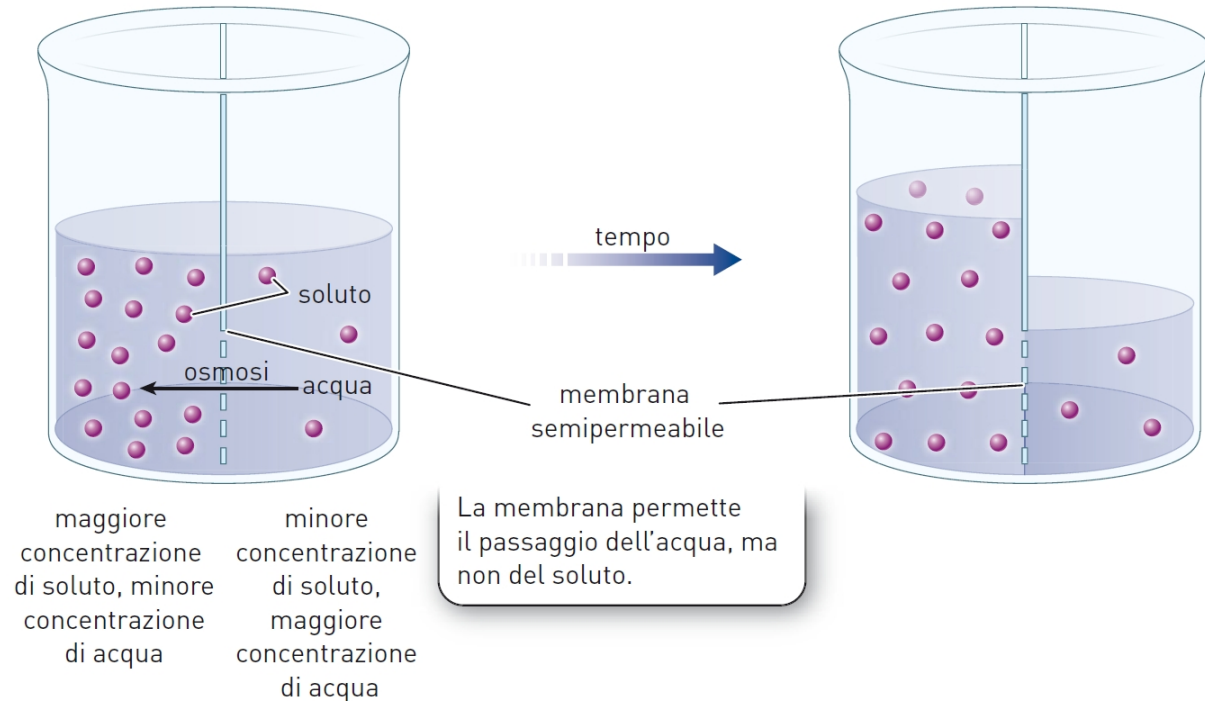
PASSIVE TRANSPORT

- MOST WATER MOVES ACROSS THE CELL MEMBRANE VIA **AQUAPORINS**.

Aquaporins: water transporters (channel proteins).

- The net movement of a solvent like water across a cell membrane is called **osmosis**.
- *Osmosis*: net movement of water **across a selectively permeable membrane** from regions of lower solute concentration (higher water concentration) to regions of higher solute concentration (lower water concentration), until equilibrium is reached.

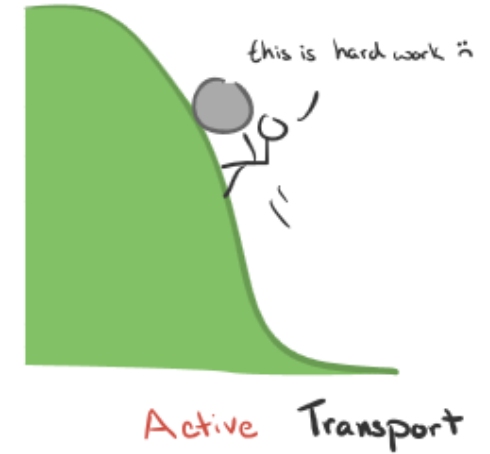
FIGURA 5.11 L'osmosi. L'osmosi è il movimento netto di un solvente, come l'acqua, attraverso una membrana semipermeabile da un'area a più bassa concentrazione di soluto a un'area a concentrazione di soluto maggiore.



THE NET MOVEMENT OF WATER CONTINUES UNTIL THE CONCENTRATION GRADIENT IS ELIMINATED OR IT IS COUNTERED BY A PRESSURE CALLED **OSMOTIC PRESSURE**.
(*OSMOTIC PRESSURE = THE PRESSURE REQUIRED TO STOP OSMOSIS.*)

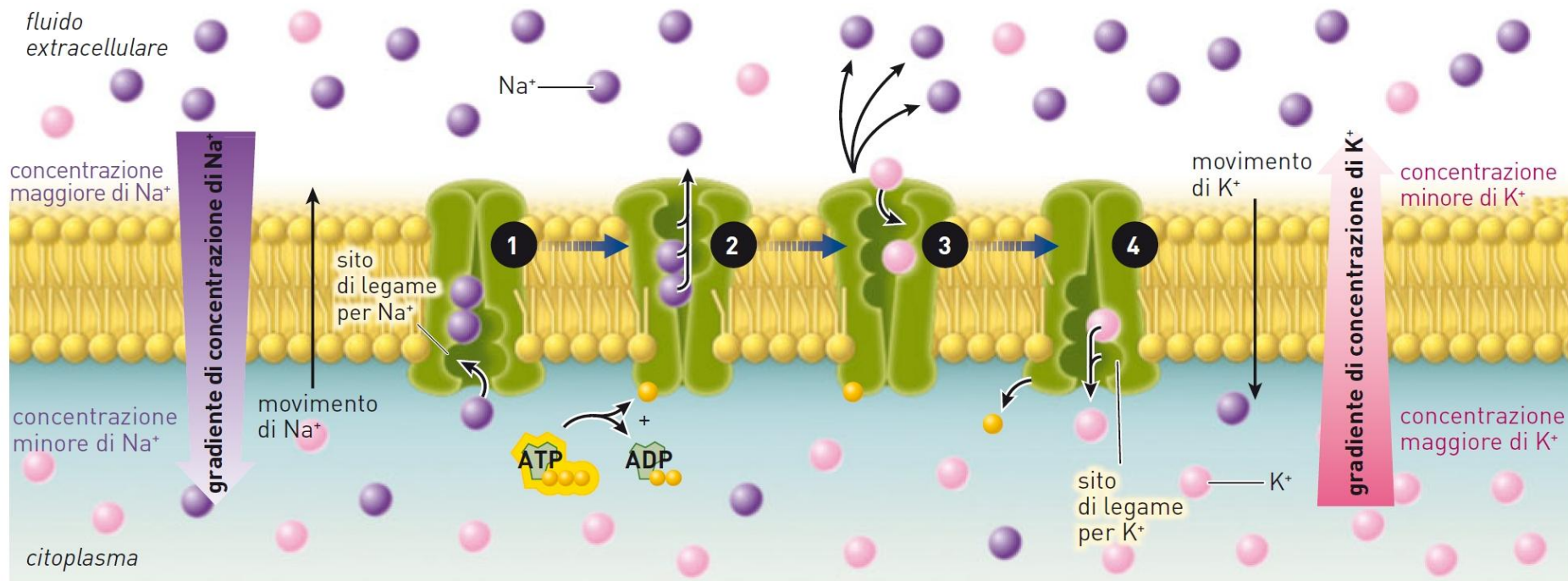
ACTIVE TRANSPORT

- **PRIMARY ACTIVE TRANSPORT (USES THE ENERGY OF ATP).**
- **SECONDARY ACTIVE TRANSPORT (USES AN ELECTROCHEMICAL GRADIENT).**



Active Transport

FIGURA 5.12 Il trasporto attivo primario. La pompa sodio-potassio è una proteina di membrana che utilizza l'energia immagazzinata nell'ATP per spostare gli ioni sodio e potassio contro i loro gradienti di concentrazione.



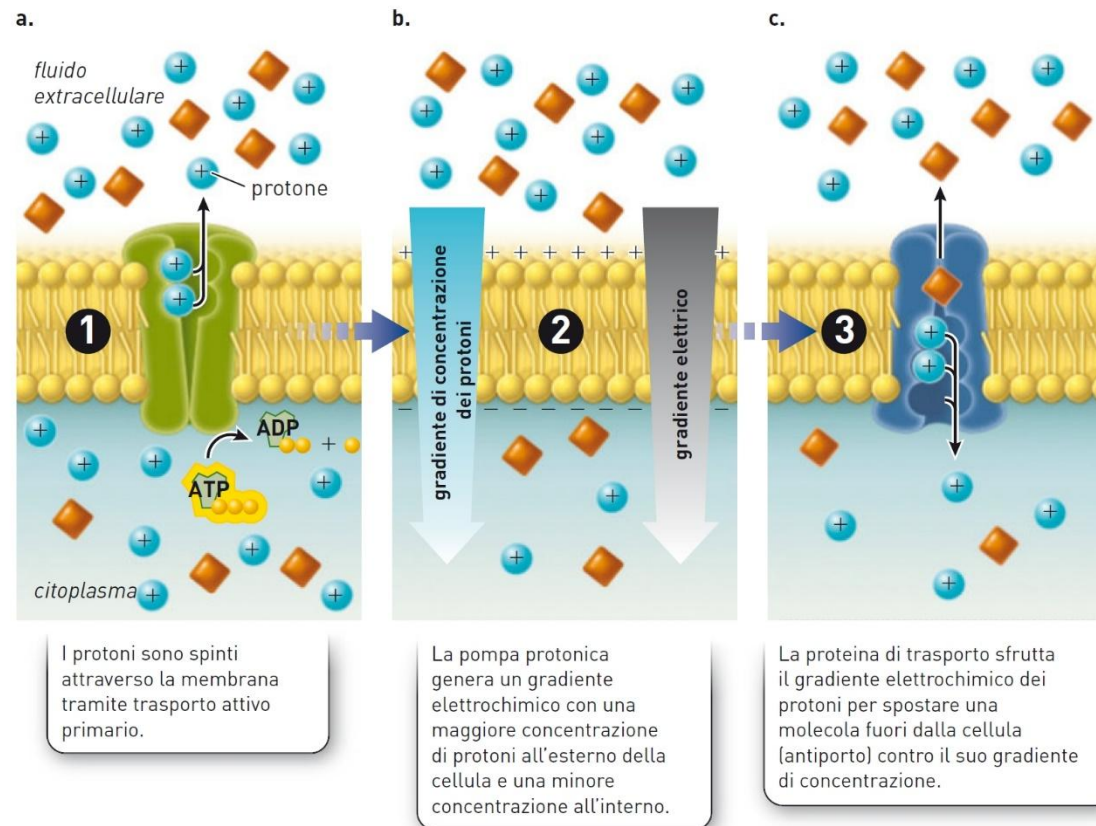
TRASPORTO ATTIVO

- **PRIMARY ACTIVE TRANSPORT (USES THE ENERGY OF ATP).**
- **SECONDARY ACTIVE TRANSPORT (USES AN ELECTROCHEMICAL GRADIENT).**



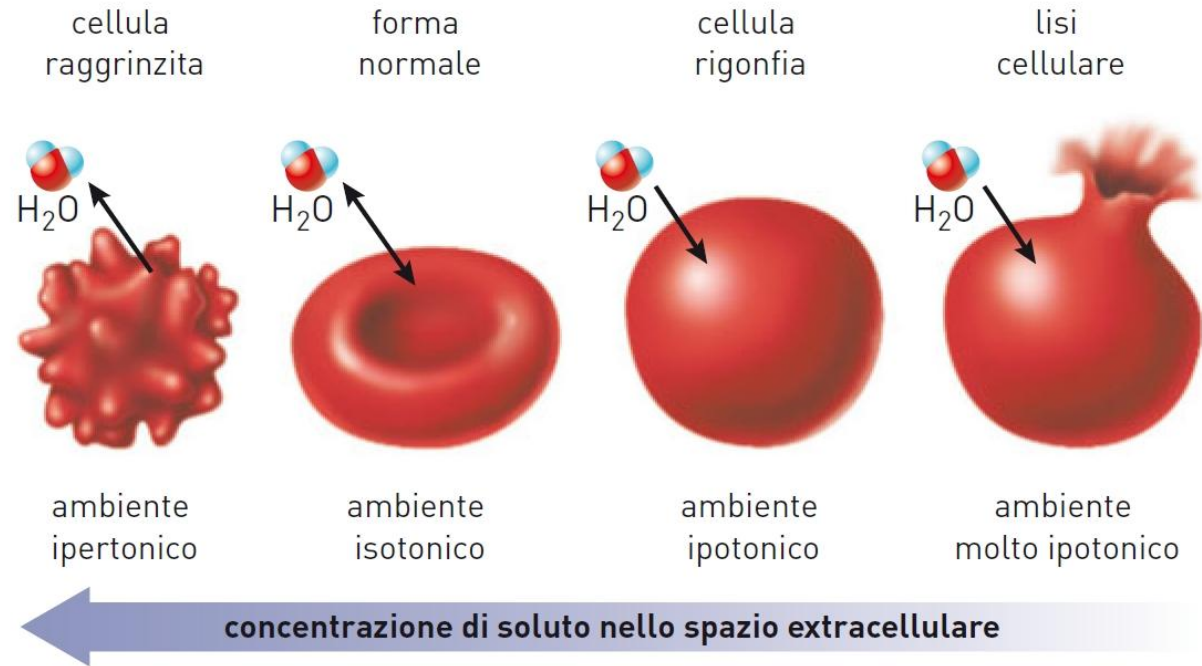
Active Transport

FIGURA 5.13 Il trasporto attivo secondario. I protoni vengono pompati attraverso una membrana (a) mediante trasporto attivo primario, generando (b) un gradiente elettrochimico che guida (c) il movimento di un'altra molecola contro il suo gradiente di concentrazione.



**CELLS MAINTAIN
THEIR SIZE (VOLUME)
AND INTERNAL
COMPOSITION
THROUGH ACTIVE
TRANSPORT.**

FIGURA 5.14 Cambiamenti nella forma dei globuli rossi a causa dell'osmosi. I globuli rossi raggrinziscono, si gonfiano o esplodono a causa del movimento netto di acqua dovuto alle differenze nella concentrazione di soluto tra l'interno e l'esterno della cellula.



ANIMAL CELLS ADDRESS THE PROBLEM THAT CAN ARISE FROM NORMAL (PASSIVE) OSMOTIC PROCESSES BY KEEPING THE INTRACELLULAR FLUID ISOTONIC WITH THE EXTRACELLULAR FLUID. THEY THEREFORE USE ACTIVE ION TRANSPORT TO MAINTAIN EQUAL CONCENTRATIONS

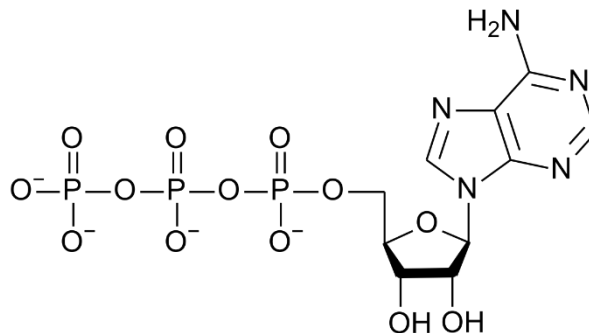
FOR EXAMPLE, THE SODIUM–POTASSIUM PUMP (Na⁺/K⁺ PUMP).

CARATTERISTICHE ESSENZIALI DI UNA CELLULA:

3) Metabolism converts energy from the environment into a form that cells can use.

METABOLISM: THE SET OF CHEMICAL REACTIONS BY WHICH CELLS CONVERT ENERGY FROM ONE FORM TO ANOTHER AND BUILD AND BREAK DOWN MOLECULES.

- ABILITY TO HARNESS ENERGY FROM THE ENVIRONMENT.
- ORGANISMS OBTAIN ENERGY FROM ONLY TWO SOURCES: SUNLIGHT AND CHEMICAL COMPOUNDS (*PHOTOTROPHS* VS. *CHEMOTROPHS*).

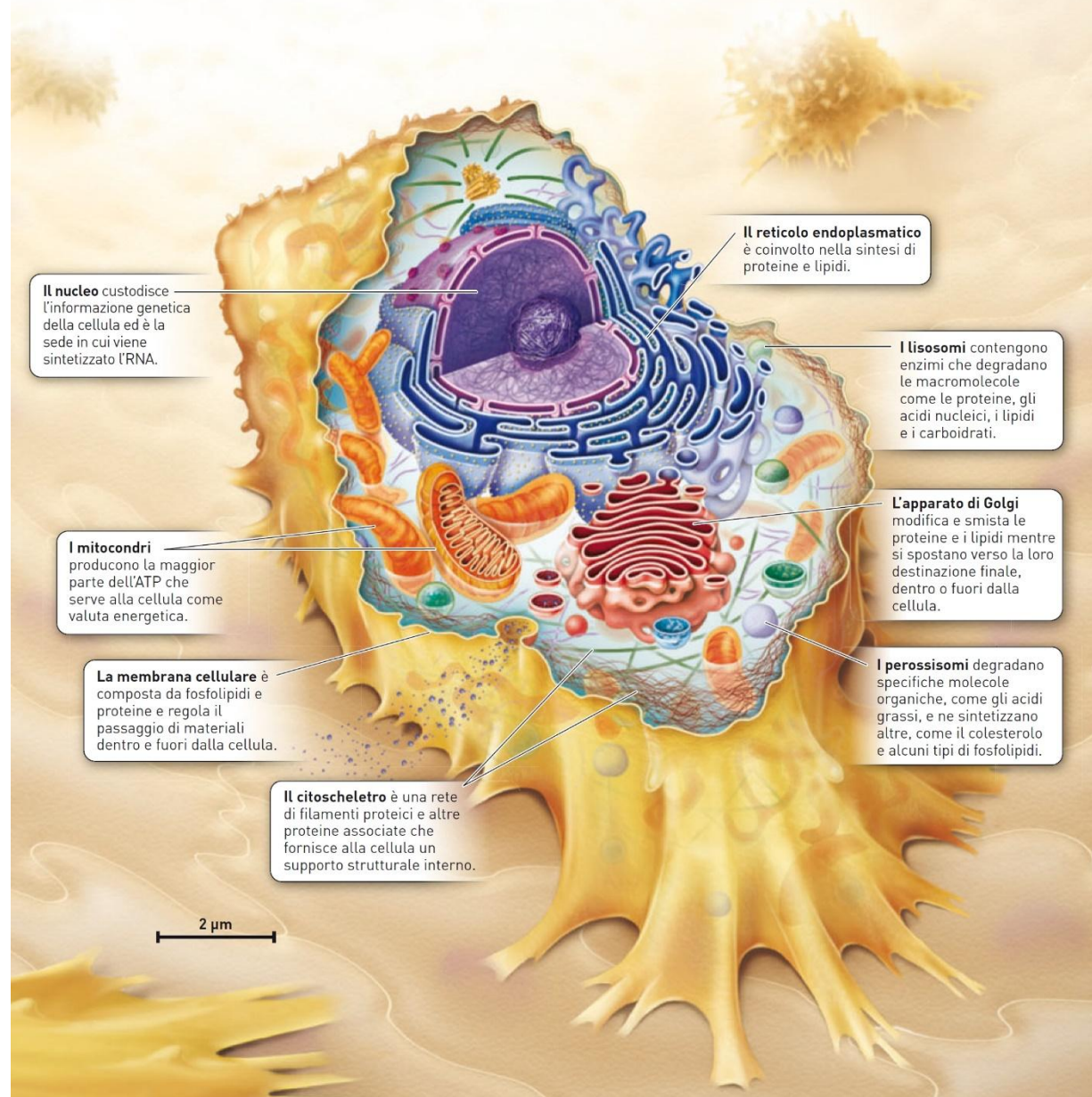


ADENOSINA TRIFOSFATO O ATP

A EUKARYOTIC CELL HAS A NUCLEUS AND SPECIALIZED INTERNAL STRUCTURES (ORGANELLES).

FIGURA 5.17 Una cellula animale e una cellula vegetale. Le cellule animali e le cellule vegetali hanno molti componenti cellulari in comune.

a. struttura tipica di una cellula animale



THE ENDOMEMBRANE SYSTEM PARTITIONS THE CELL INTO COMPARTMENTS (ORGANELLES):

NUCLEUS

FIGURA 5.18 Il sistema di endomembrane. Il sistema di endomembrane è una serie di compartimenti della cellula eucariote, delimitati da membrane.

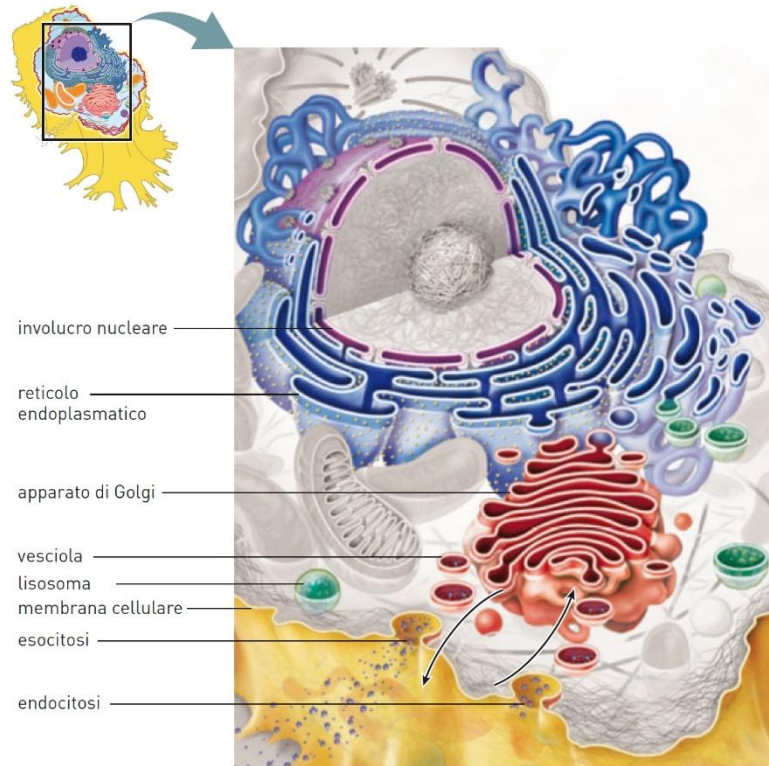
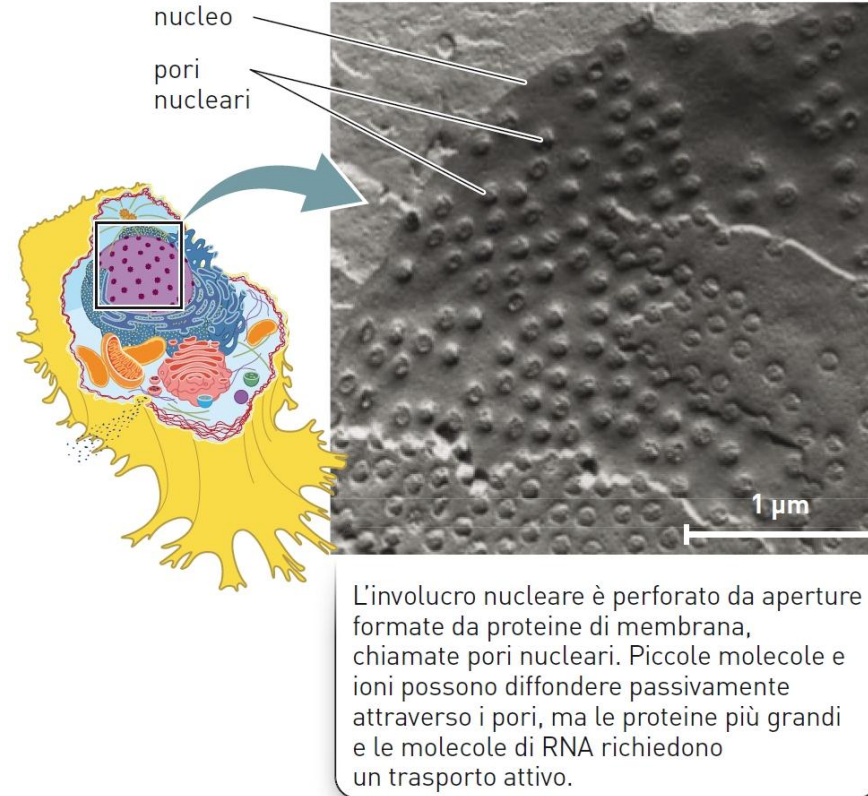


FIGURA 5.19 Visione superficiale dell'involucro nucleare. Il nucleo è circondato da una doppia membrana e ospita il DNA cellulare.



- ENCLOSED BY A DOUBLE MEMBRANE.
- PROTEINS AND LARGE MOLECULES SUCH AS MRNA REQUIRE ACTIVE TRANSPORT (THROUGH NUCLEAR PORE COMPLEXES).

THE ENDOMEMBRANE SYSTEM PARTITIONS THE CELL INTO COMPARTMENTS (ORGANELLES):

THE ENDOPLASMIC RETICULUM (ER)

FIGURA 5.18 Il sistema di endomembrane. Il sistema di endomembrane è una serie di compartimenti della cellula eucariote, delimitati da membrane.

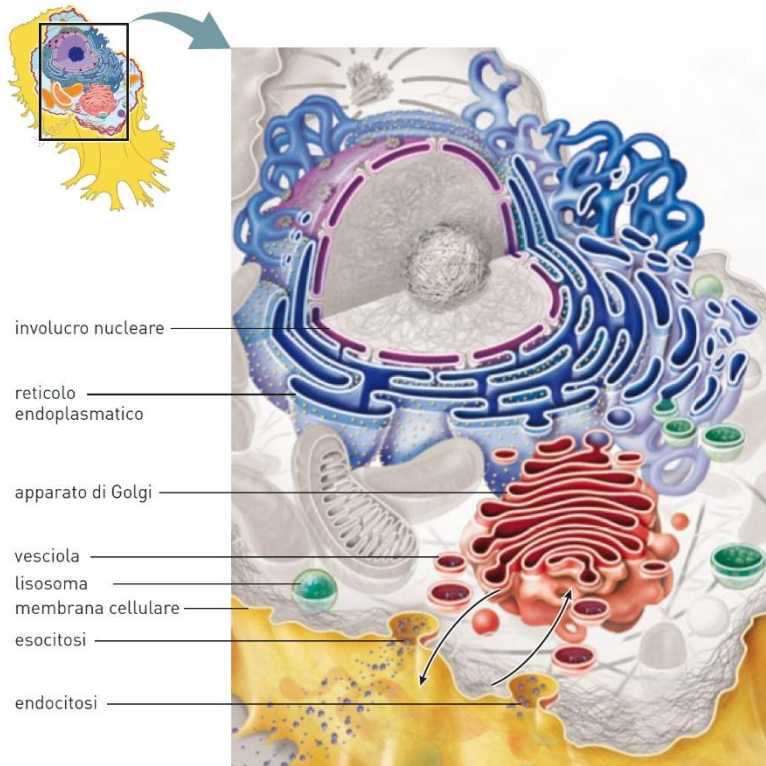
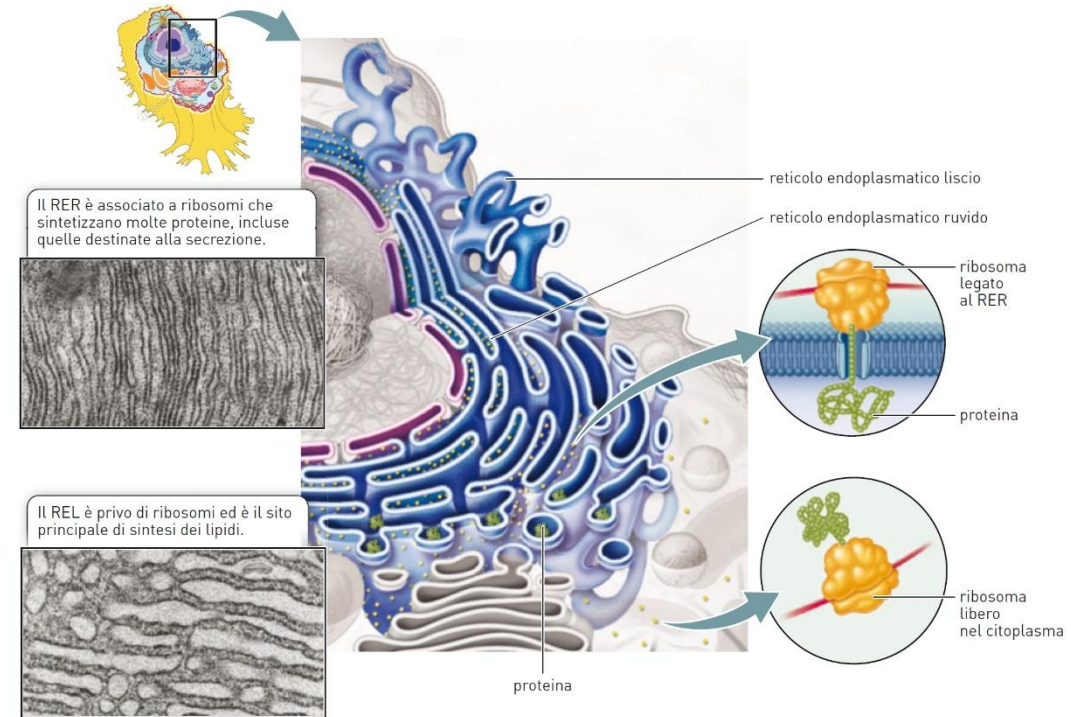


FIGURA 5.20 Il reticolo endoplasmatico (RE). Il reticolo endoplasmatico è un sito importante per la sintesi proteica e lipidica. Le proteine vengono sintetizzate anche nel citoplasma.



- BOUNDED BY A SINGLE MEMBRANE.
- COMPLEX NETWORK OF TUBULES.
- THE INTERNAL SPACE IS CALLED THE LUMEN.
- FUNCTIONS (SYNTHESIS):
 PROTEINS — ROUGH ER (RER)
 LIPIDS — SMOOTH ER (SER)

THE ENDOMEMBRANE SYSTEM PARTITIONS THE CELL INTO COMPARTMENTS (ORGANELLES):

GOLGI APPARATUS

FIGURA 5.18 Il sistema di endomembrane. Il sistema di endomembrane è una serie di compartimenti della cellula eucariote, delimitati da membrane.

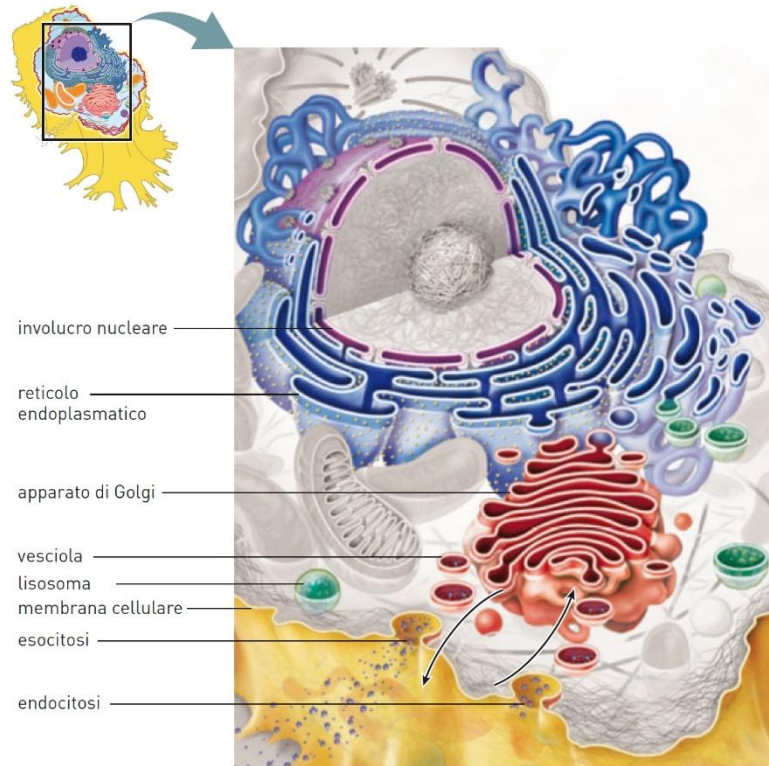
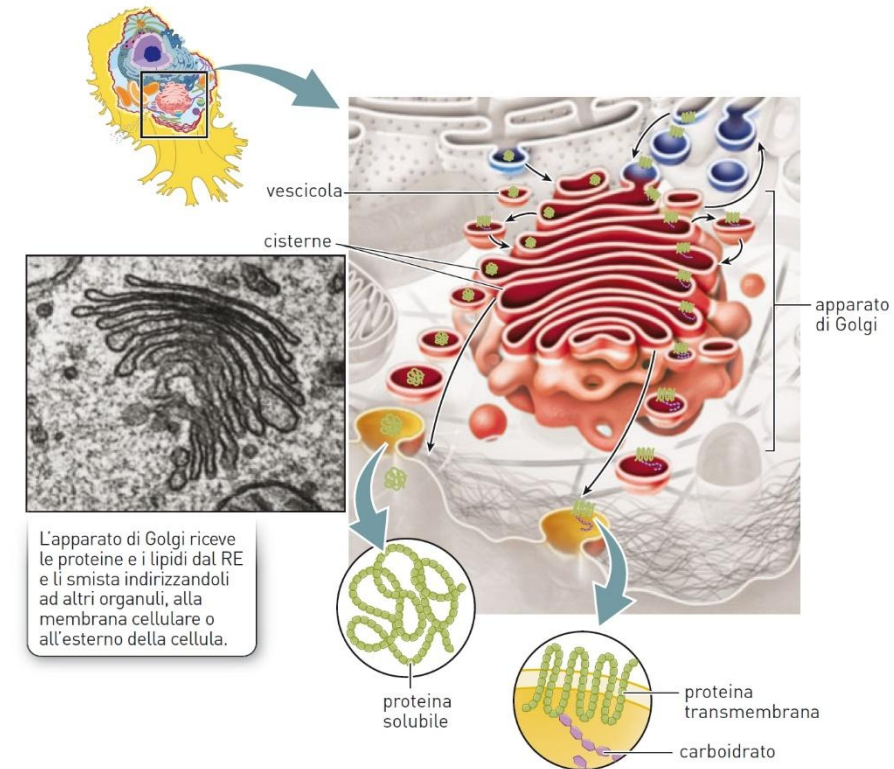


FIGURA 5.21 L'apparato di Golgi. L'apparato di Golgi smista le proteine e i lipidi direzionandoli verso altri organuli, la membrana cellulare o l'esterno della cellula.



Three main roles (Golgi apparatus):

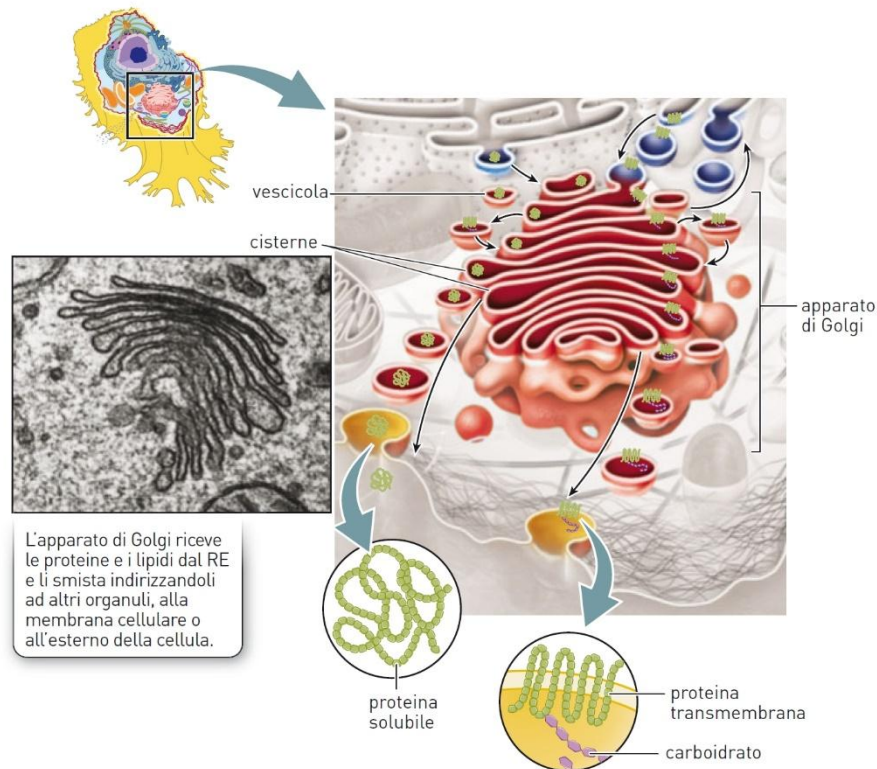
- **Further modifies** proteins and lipids produced by the **ER**.
- **Acts as a sorting station** directing proteins and lipids to their final destinations.
- **Site of synthesis** for most of the cell's **carbohydrates**.

Microscopy: appears as a series of membrane sacs called **cisternae**, surrounded by **vesicles** through which proteins and lipids are sorted to their final destinations.

THE ENDOMEMBRANE SYSTEM PARTITIONS THE CELL INTO COMPARTMENTS (ORGANELLES):

GOLGI APPARATUS

FIGURA 5.21 L'apparato di Golgi. L'apparato di Golgi smista le proteine e i lipidi direzionandoli verso altri organuli, la membrana cellulare o l'esterno della cellula.



GLYCOSYLATION: A REACTION IN WHICH SUGAR MOLECULES ARE **COVALENTLY ATTACHED** TO LIPIDS OR TO SPECIFIC AMINO-ACID RESIDUES OF CERTAIN PROTEINS, PRODUCING:

- **GLYCOLIPIDS**
 - **GLYCOPROTEINS.**
- SUGARS ATTACHED TO PROTEINS CAN ACT AS A **PROTECTIVE "CAP,"** SHIELDING THEM FROM DIGESTIVE ENZYMES.
 - THEY FORM A **FLEXIBLE, PROTECTIVE COATING** ON THE MEMBRANE (*THE GLYCOLALYX*).
 - THEY PROVIDE **SPECIFIC RECOGNITION** BY OTHER CELLS OR EXTRACELLULAR MOLECULES.
 - **EXAMPLE:** BLOOD GROUPS **A, B, AB, O** ARE DEFINED BY PARTICULAR SUGARS ON THE SURFACE OF RED BLOOD CELLS.

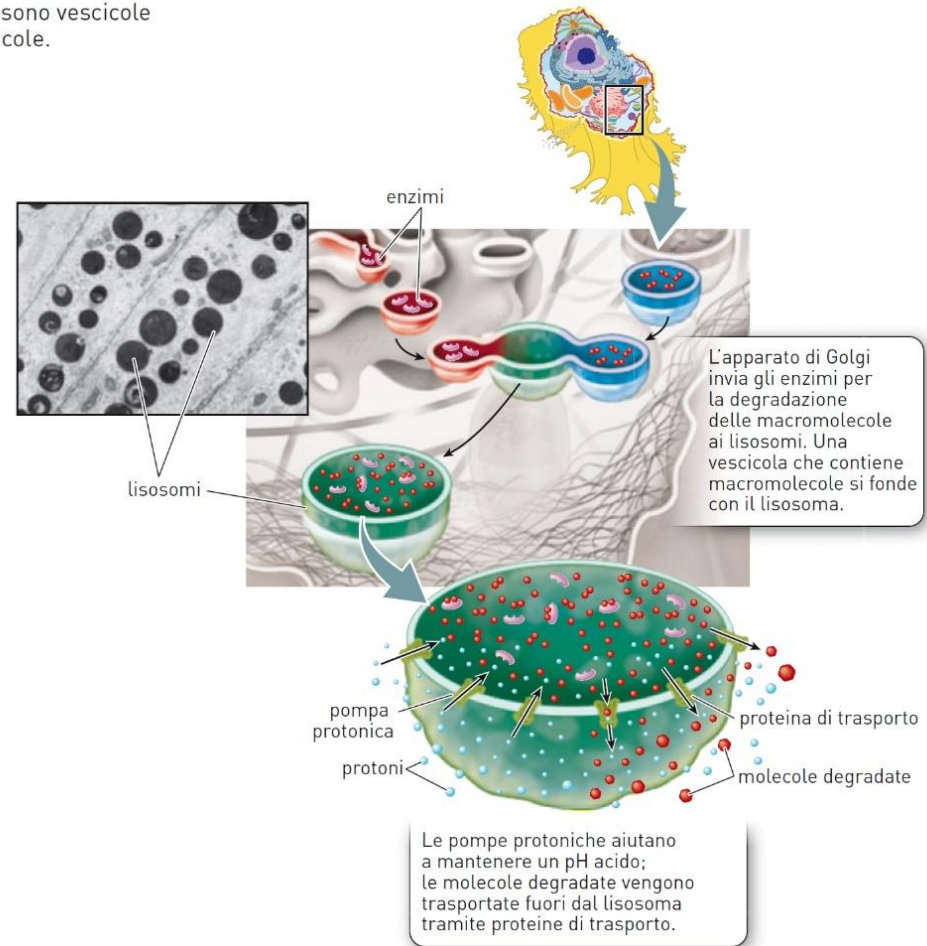
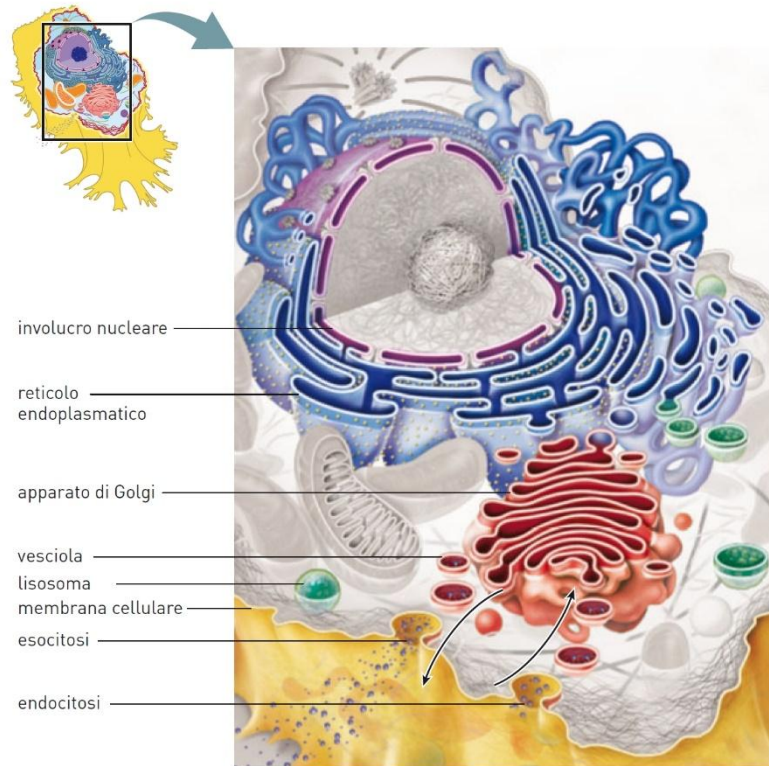
ALTHOUGH VESICLES USUALLY TRAVEL **FROM THE ER TO THE GOLGI**, THEY SOMETIMES GO **IN THE REVERSE DIRECTION** TO RETRIEVE PROTEINS SENT TO THE WRONG PLACE AND TO **RECYCLE MEMBRANE COMPONENTS.**

FIGURA 5.22 | lisosomi. I lisosomi sono vescicole che scompongono le macromolecole.

THE ENDOMEMBRANE SYSTEM PARTITIONS THE CELL INTO COMPARTMENTS (ORGANELLES):

LYSOSOMES

FIGURA 5.18 Il sistema di endomembrane. Il sistema di endomembrane è una serie di compartimenti della cellula eucariote, delimitati da membrane.



Lysosomes: specialized vesicles derived from the Golgi apparatus that degrade damaged or unnecessary macromolecules.

They contain a variety of enzymes that break down proteins, nucleic acids, lipids, and complex carbohydrates.

MITOCHONDRIAL MEMBRANES ARE NOT PART OF THE ENDOMEMBRANE SYSTEM; MITOCHONDRIA ARE SPECIALIZED ORGANELLES OF THE CELL.

MITOCHONDRIA

FIGURA 7.3 Le quattro fasi della respirazione cellulare. La respirazione cellulare è composta da glicolisi, ossidazione del piruvato, ciclo dell'acido citrico e fosforilazione ossidativa.

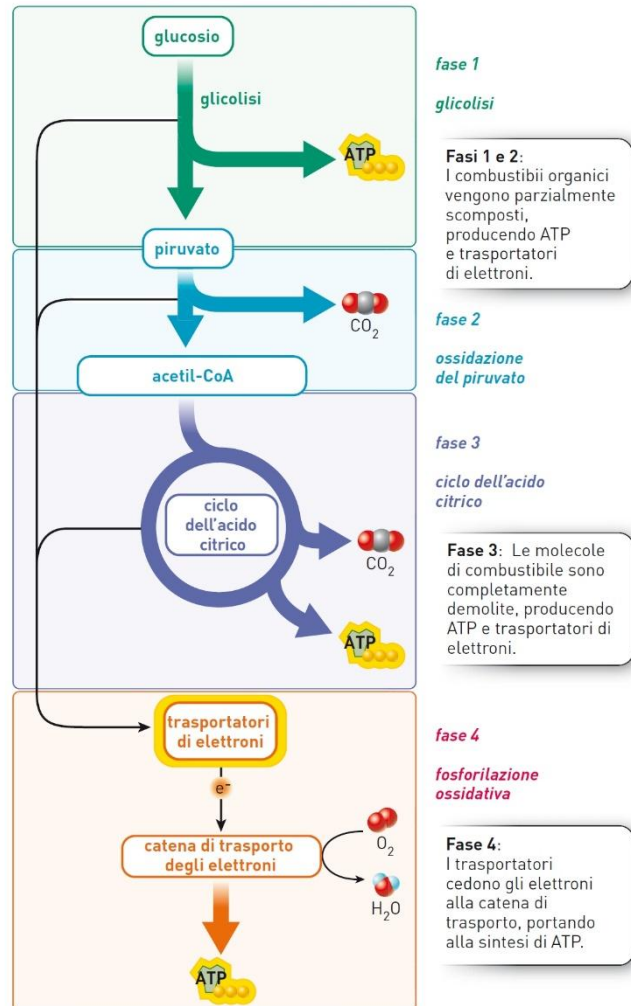
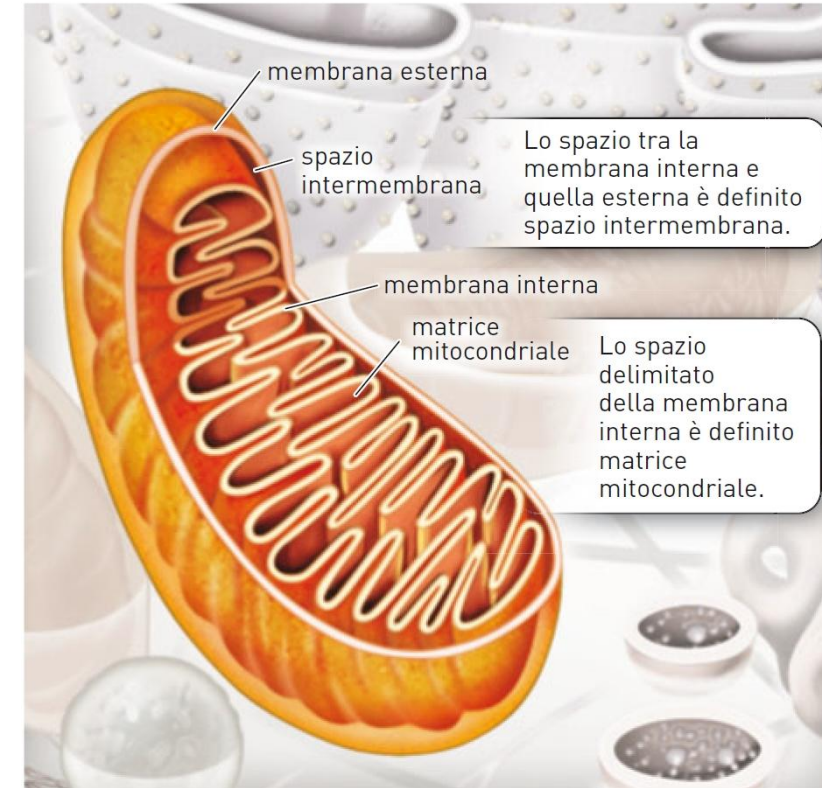


FIGURA 7.6 Le membrane e i compartimenti mitocondriali.



Double membrane—outer and inner; the inner membrane is highly folded into cristae.

Krebs cycle and cellular respiration.

They use the energy stored in sugars to convert it into ATP.

GLYCOLYSIS: THE PROCESS BY WHICH GLUCOSE IS PARTIALLY BROKEN DOWN TO PRODUCE PYRUVATE (PYRUVIC ACID).

PYRUVATE OXIDATION: OXIDATION OF PYRUVATE TO ACETYL-COA.

CITRIC ACID CYCLE (KREBS CYCLE): A SERIES OF REACTIONS THAT COMPLETELY OXIDIZE THE FUEL, PRODUCING ATP AND ELECTRON CARRIERS.

OXIDATIVE PHOSPHORYLATION: A SERIES OF REACTIONS IN WHICH ELECTRON CARRIERS DONATE ELECTRONS, LEADING TO THE PRODUCTION OF LARGE AMOUNTS OF ATP.

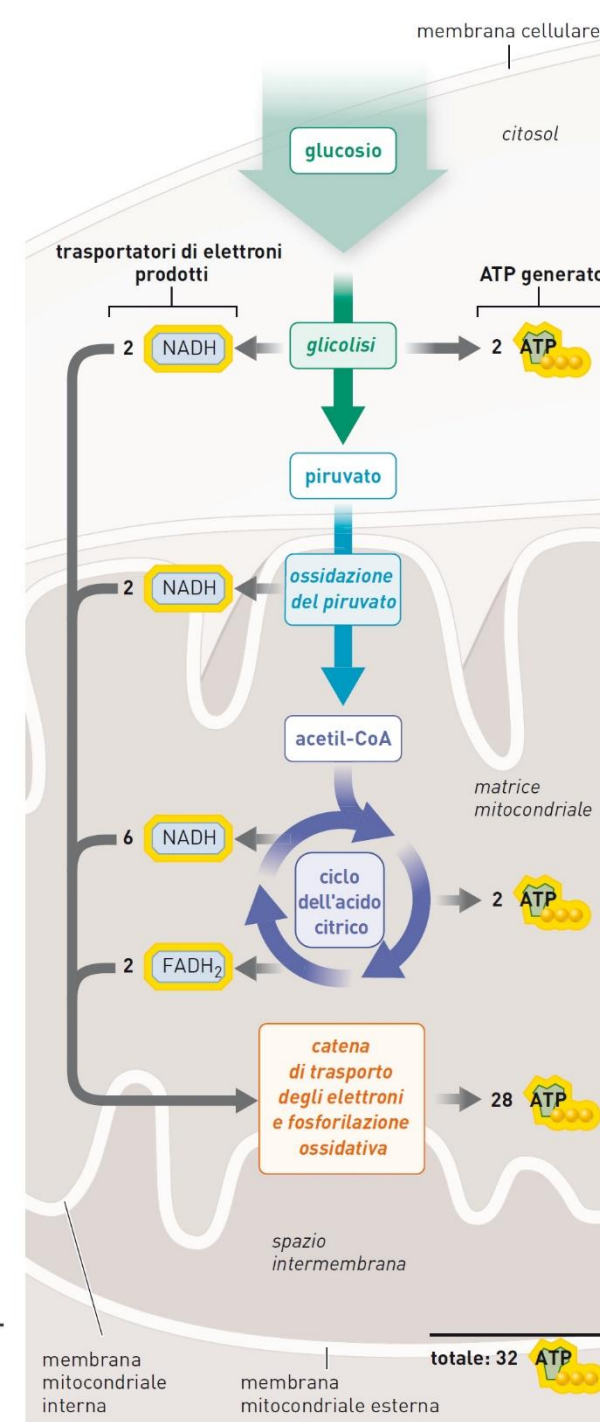


FIGURA 7.13 Il flusso dell'energia nella respirazione cellulare. Da una singola molecola di glucosio vengono prodotte 32 molecole di ATP.

FIGURA 7.9 Le molecole prodotte a partire dal ciclo dell'acido citrico. Molte molecole organiche possono essere sintetizzate a partire dagli intermedi del ciclo dell'acido citrico.

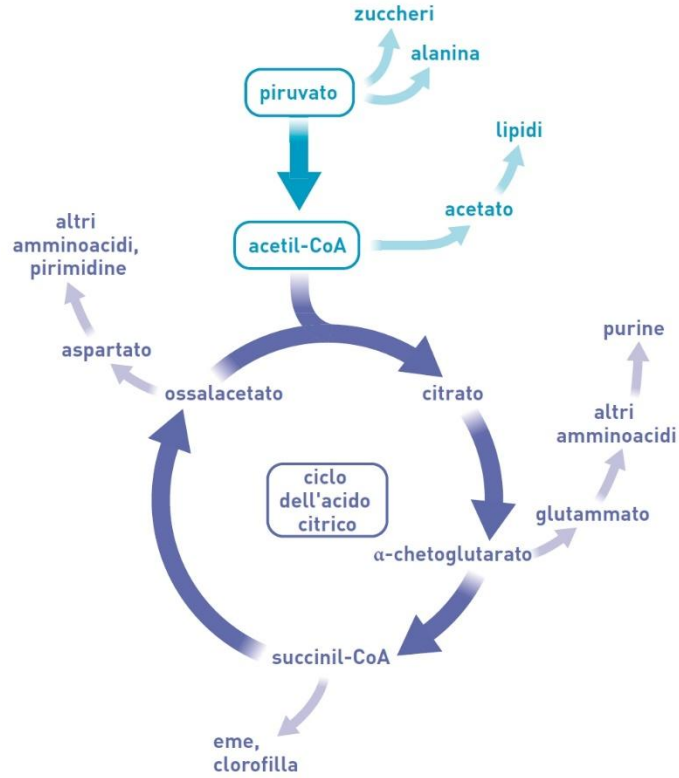
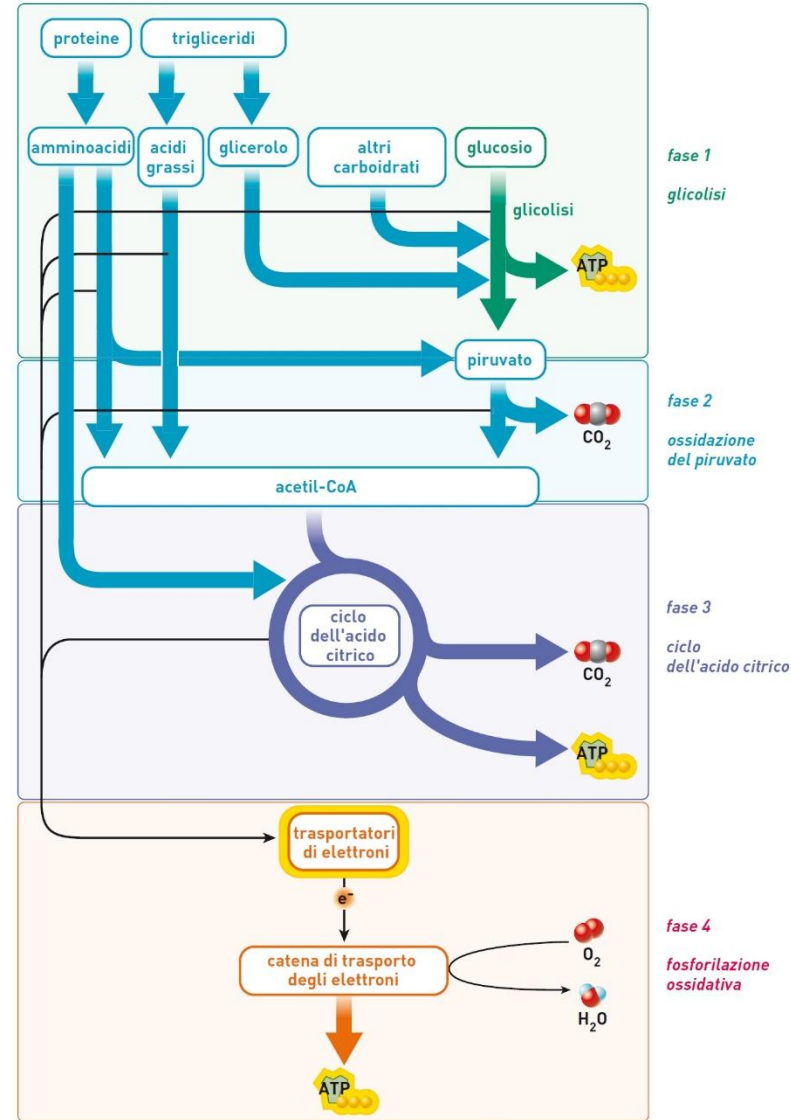


FIGURA 7.18 Gli altri combustibili organici. Gli zuccheri diversi dal glucosio, i lipidi, come i trigliceridi, e le proteine sono tutte molecole di combustibile che vengono demolite ed entrano nelle vie metaboliche della respirazione cellulare in punti diversi.



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